



UNIVERSITEIT VAN AMSTERDAM

# Forensic Probability and Statistics

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INTRODUCTION. This course aims to understand the basics of the theory and philosophy of forensic statistics and probability, and to apply these to concrete situations. The course is a natural mixture of mainly rigorous probabilistic and statistical theory on the one hand, and elements from biology, forensic science, legal science, and philosophy on the other.

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# Some Philosophy of Probability, Statistics, and Forensic Science

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# Evidence and the Likelihood Ratio

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## SECTION 2.1

## | From Evidence to Likelihood Ratio

Taking the existence of a *prior assessment* for granted, we might say that *data*  $E$  provides evidence (against or in favour of) a *hypothesis*  $H$  if

$$\mathbb{P}(H | E) = \frac{\mathbb{P}(E \cdot H)}{\mathbb{P}(E)} = \mathbb{P}(H) \cdot \frac{\mathbb{P}(E | H)}{\mathbb{P}(E)} \neq \mathbb{P}(H). \quad (2.1)$$

DEFINITION 2.1.

We define the *likelihood ratio*  $\text{LR}_{H_1, H_2}(E)$  of  $H_1$  versus  $H_2$  as

$$\text{LR}_{H_1, H_2}(E) := \frac{\mathbb{P}(E | H_1)}{\mathbb{P}(E | H_2)}.$$

And we have the *odds form of Bayes' rule*:

$$\underbrace{\frac{\mathbb{P}(H_1 | E)}{\mathbb{P}(H_2 | E)}}_{\text{posterior odds}} = \text{LR}_{H_1, H_2}(E) \times \underbrace{\frac{\mathbb{P}(H_1)}{\mathbb{P}(H_2)}}_{\text{prior odds}}. \quad (2.2)$$

EXAMPLE 2.2. Suppose that we have a vase containing 100 balls. Each ball is either black or white. Consider the hypotheses:

- $H_1$ : The vase contains 90 white balls and 10 black balls.
- $H_2$ : The vase contains 50 white and 50 black balls.

Suppose we draw 10 balls with replacement.

- $E$ : Each of the 10 drawn balls turns out to be white.

We have

$$\mathbb{P}(E | H_1) = \left(\frac{9}{10}\right)^{10}, \quad \mathbb{P}(E | H_2) = \left(\frac{1}{2}\right)^{10}.$$

The corresponding likelihood ratio is easily computed to be  $\left(\frac{9}{5}\right)^{10}$ , which is roughly 357. Consider hypothesis

- $H_3$ : The vase contains only white balls.

Suppose that in addition, the prior probabilities for each of the three hypotheses  $H_1, H_2$ , and  $H_3$  are undisputed and equal to  $\frac{1}{3}$ . It is now straightforward to compute

$$\mathbb{P}(H_1 | E) = 0.258, \quad \mathbb{P}(H_2 | E) = 0.001, \quad \mathbb{P}(H_3 | E) = 0.741.$$

EXAMPLE 2.3. Consider a deck of 52 cards, where each card has a number between 1 and 52 (inclusive).<sup>1</sup>

- $H_n$ : The deck is a normal deck of cards, where each number between 1 and 52 is used once.
- $H_u$ : All cards in the deck have the same (unspecified) number.
- $H_i$  ( $i = 1, \dots, 52$ ): All cards have number  $i$ .
- $E$ : Shuffle the deck, draw one card, and the result is the number 12.

We can easily see that

$$LR_{H_{12}, H_n} = \frac{\mathbb{P}(E | H_{12})}{\mathbb{P}(E | H_n)} = \frac{1}{\frac{1}{52}} = 52.$$

The hypothesis  $H_{12}$  is an example of a so-called *data-driven hypothesis*, that is, a hypothesis that one formulates only after obtaining the evidence.

We can also compute  $LR_{H_n, H_u}$ , the likelihood ratio of two hypotheses which are not data driven. If we make the additional assumption that  $\mathbb{P}(H_i | H_u) = \frac{1}{52}$  for all  $i$ , then we have

$$\mathbb{P}(E | H_u) = \sum_{i=1}^{52} \mathbb{P}(E | H_i) \mathbb{P}(H_i | H_u) = \frac{1}{52} \sum_{i=1}^{52} \mathbb{P}(E | H_i) = \frac{1}{52}.$$

Hence it follows that  $LR_{H_n, H_u}(E) = 1$ , so the evidence of obtaining a 12 gives no information whatsoever about the deck being normal or uniform.

EXAMPLE 2.4 (The Island Problem). Our starting point is an island. On this island a crime is committed, and we assume with certainty that the criminal  $C$  is an inhabitant of the island. We denote the collection of all people on the island by  $X$ . We introduce indicator random variables  $\Gamma_x$ , taking value 1 if  $x \in X$  has a certain characteristic  $\Gamma$  and 0 otherwise. We assume that the  $\Gamma_x$  are *i.i.d.* Bernoulli distributed with success probability  $p$ .

Now we obtain information that  $\Gamma$  has been found at the scene of the crime, and we assume that we may conclude that  $\Gamma_C = 1$ . Suppose further that we obtain the information that a certain identified person  $S$  has  $\Gamma$ . Writing  $E_C := \{\Gamma_C = 1\}$ ,  $G := \{S = C\}$ , and  $E_S := \{\Gamma_S = 1\}$ , we are interested in the likelihood ratio given by

$$\frac{\mathbb{P}(E_C, E_S | G)}{\mathbb{P}(E_C, E_S | G^c)} = \frac{p}{p^2} = \frac{1}{p}, \quad \frac{\mathbb{P}(E_C | G, E_S)}{\mathbb{P}(E_C | G^c, E_S)} = \frac{1}{p}.$$

If we write  $\pi_S := \mathbb{P}(G)$  for the prior probability of guilt, we obtain

$$\frac{\mathbb{P}(G | E_C, E_S)}{\mathbb{P}(G^c | E_C, E_S)} = \frac{\mathbb{P}(E_C | G, E_S)}{\mathbb{P}(E_C | G^c, E_S)} \cdot \frac{\mathbb{P}(G | E_S)}{\mathbb{P}(G^c | E_S)} = \frac{1}{p} \cdot \frac{\pi_S}{1 - \pi_S}.$$

<sup>1</sup> We use numbers instead of the usual card notation for notational convenience.

Since we also have that  $\mathbb{P}(G \mid E_C, E_S) + \mathbb{P}(G^c \mid E_C, E_S) = 1$ , it now follows that the posterior probability of guilt is given by

$$\mathbb{P}(G \mid E_C, E_S) = \frac{1}{1 + \frac{p(1-\pi_S)}{\pi_S}}.$$

SECTION 2.2

# The Distribution of the Likelihood Ratio

We write the probability of finding a likelihood ratio of  $x$  when we sample  $E$  according to the distribution corresponding to  $H_2$  as  $\mathbb{P}(\text{LR}_{H_1, H_2} = x \mid H_2)$ .

## Basic Properties of the Likelihood Ratio Distribution

PROPOSITION 2.5.

We have

$$\mathbb{P}(\text{LR} = x \mid H_1) = x\mathbb{P}(\text{LR} = x \mid H_2), \quad \forall 0 \leq x < \infty. \quad (2.3)$$

*Proof.* Let

$$\mathcal{E}_x = \left\{ E : \text{LR}(E) = \frac{\mathbb{P}(E \mid H_1)}{\mathbb{P}(E \mid H_2)} = x \right\}$$

be the set of all pieces of evidence for which we obtain a likelihood ratio equal to  $x$ . Then

$$\mathbb{P}(\text{LR} = x \mid H_1) = \sum_{E \in \mathcal{E}_x} \mathbb{P}(E \mid H_1) = \sum_{E \in \mathcal{E}_x} x\mathbb{P}(E \mid H_2) = x\mathbb{P}(\text{LR} = x \mid H_2),$$

which is what we needed to show. Note that it is the second equality in which we use that  $x < \infty$ .  $\square$

It follows that if  $\mathbb{P}(\text{LR} = x \mid H_2) > 0$ , then

$$\frac{\mathbb{P}(\text{LR} = x \mid H_1)}{\mathbb{P}(\text{LR} = x \mid H_2)} = x.$$

Since  $\mathbb{P}(\text{LR} = \infty \mid H_2) = 0$ , for all  $k = 0, 1, 2, \dots$ , we can write

$$\begin{aligned} \mathbb{E}[\text{LR}^{k+1} \mid H_2] &= \sum_{0 \leq x < \infty} x^{k+1} \mathbb{P}(\text{LR} = x \mid H_2) \\ &= \sum_{0 \leq x < \infty} x^k \mathbb{P}(\text{LR} = x \mid H_1) \leq \mathbb{E}[\text{LR}^k \mid H_1], \end{aligned}$$

with equality if

$$\mathbb{P}(\text{LR} = \infty \mid H_1) = 0. \quad (2.4)$$

It is easy to see that (2.4) is equivalent to

$$\mathbb{P}(E \mid H_1) > 0 \implies \mathbb{P}(E \mid H_2) > 0, \quad \forall E. \quad (2.5)$$

In words, (2.5) says that if  $H_1$  is able to explain  $E$ , so is  $H_2$ . Assuming (2.5), we thus have

$$\mathbb{E}[\text{LR}^{k+1} \mid H_2] = \mathbb{E}[\text{LR}^k \mid H_1].$$

Taking  $k = 0$  leads to the remarkable property that

$$\mathbb{E}[\text{LR} \mid H_2] = 1.$$

Taking  $k = 1$  leads to

$$\mathbb{E}[\text{LR} \mid H_1] = \mathbb{E}[\text{LR}^2 \mid H_2] \geq (\mathbb{E}[\text{LR} \mid H_2])^2 = 1.$$

## Misleading Evidence

PROPOSITION 2.6.

For all  $x > 0$  we have

$$\mathbb{P}(\text{LR} \geq x \mid H_2) \leq \frac{1}{x}$$

by Markov's inequality since  $\mathbb{E}[\text{LR} \mid H_2] = 1$ .

## The Weight of the Evidence

It turns out that indeed if we look on a logarithmic scale, we get somewhat better behaviour. We have

$$\mathbb{E}[\log \text{LR} \mid H_2] = \sum_{x \geq 0} \log x \mathbb{P}(\text{LR} = x \mid H_2),$$

where the sum is over all  $x$  that can be taken by LR, and where any base is allowed for the logarithm. If  $P(\text{LR} = 0 \mid H_2) > 0$  this expression is equal to  $-\infty$ . Otherwise by *Jensen's inequality* we have

$$\mathbb{E}[\log \text{LR} \mid H_2] \leq \log \mathbb{E}[\text{LR} \mid H_2] = 0.$$

For likelihood ratios encountered when  $H_1$  is true, we have

$$\begin{aligned} \mathbb{E}[\log \text{LR} \mid H_1] &= \sum_{x > 0} \log x \mathbb{P}(\text{LR} = x \mid H_1) \\ &= \sum_{x > 0} x \log x \mathbb{P}(\text{LR} = x \mid H_2) \\ &\geq \mathbb{E}[\text{LR} \mid H_2] \log \mathbb{E}[\text{LR} \mid H_2] = 0, \end{aligned}$$

where we have used that  $f(x) = x \log x$  is a convex function and hence again by *Jensen's inequality*,  $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$  for any random variable  $X$ .

EXAMPLE 2.7 (Canonical Experiment). Consider two urns, one containing only white balls, and the other containing an equal number of white and black balls. We choose one urn and draw a succession of balls from it, with replacement. We have

- $H_1$ : All white.
- $H_2$ : Half white and half black.
- $E$ : The colours of the chosen balls.

Suppose now that you draw five balls and they are all white. The likelihood ratio of  $H_1$  versus  $H_2$  corresponding to this outcome is 32. A likelihood ratio of 357 as occurring in [Example 2.2](#) corresponds to drawing between 8 and 9 white balls in a row. The number of white balls that must be chosen in order for the likelihood ratio to reach a certain level  $L$  is  $\log_2 L$ .

## Exercises

EXERCISE 2.1. Suppose that at the scene of a crime, a DNA profile has been obtained which we may assume is donated by the criminal  $C$ . There are two suspects  $S_1$  and  $S_2$  who both have this particular DNA profile, and we write:

- $H_1$ :  $C$  is one of them.
- $H_2$ : The unknown  $C$  is unrelated to both  $S_1$  and  $S_2$ .

We are interested in the likelihood ratio of the findings. In an arbitrarily large population, the population frequency of this profile will have a certain value  $\beta$ .

- (i) Explain that if we knew the value  $\beta$ , then the likelihood ratio would be  $\frac{1}{\beta}$ .
- (ii) Suppose that our knowledge about  $\beta$  is described by the random variable  $W$  which is uniformly distributed on  $[0, a]$ , for some  $0 < a \leq 1$ . Show that the likelihood ratio of the findings as a function of  $a$  is equal to  $\frac{4}{3a}$ .
- (iii) Explain why it is reasonable that the likelihood ratio approaches infinity for  $a \rightarrow 0$ , and is decreasing in  $a$ .

SOLUTION. (i) Let the DNA profile be  $g$  and let  $\{G_i = g\}$  be the event that  $S_i$  has profile  $g$ . Let  $\{G_C = g\}$  be the event that the trace donor has this profile too. Then

$$\text{LR} = \frac{\mathbb{P}(G_1 = g, G_2 = g, G_C = g \mid H_1)}{\mathbb{P}(G_1 = g, G_2 = g, G_C = g \mid H_2)} = \frac{\beta^2}{\beta^3} = \frac{1}{\beta}.$$

(ii) Now,

$$\begin{aligned}\mathbb{P}(G_1 = g, G_2 = g, G_C = g \mid H_1) &= \frac{1}{a} \int_0^a x^2 dx = \frac{1}{3}a^3, \\ \mathbb{P}(G_1 = g, G_2 = g, G_C = g \mid H_2) &= \frac{1}{a} \int_0^a x^3 dx = \frac{1}{4}a^4.\end{aligned}$$

$$\text{So LR} = \frac{4}{3a}.$$

- (iii) A smaller value of  $a$  corresponds to greater certainty that the profile is rare, with a frequency of at most  $a$ . As  $a \rightarrow 0$ , the evidence proves  $H_1$  in the limit. As  $a$  grows, we become more uncertain about the frequency of the profile, with  $a = 1$  corresponding to maximal uncertainty.

□

EXERCISE 2.2. Consider an autosomal locus with two alleles  $a$  and  $b$ , with frequencies  $p_a > 0$  and  $p_b > 0$ , respectively. Let  $G_{AF}$  and  $G_C$  denote the genotypes on this locus for an alleged father  $AF$  and child  $C$ , respectively. Let

- $H_1$ :  $AF$  is the father of  $C$ .
- $H_2$ :  $AF$  is unrelated to  $C$ .

We denote

$$\text{LR}_{H_1, H_2}(G_{AF}, G_C) = \frac{\mathbb{P}(G_{AF}, G_C \mid H_1)}{\mathbb{P}(G_{AF}, G_C \mid H_2)}.$$

- (i) Tabulate the possible  $\mathbb{P}(G_{AF} = g_1, G_C = g_2 \mid H_1)$  for all possible genotypes  $g_1, g_2$  that parents and child may have.

(ii) Show that, regardless of allele frequencies  $p_a$  and  $p_b$ , we have

$$\mathbb{E}[\text{LR}_{H_1, H_2}(G_{AF}, G_C) \mid H_1] = \frac{5}{4}.$$

(iii) Let  $H_3$  denote the hypothesis that  $AF$  and  $C$  are monozygotic twins. Compute  $\mathbb{E}[\text{LR}_{H_3, H_2}(G_{AF}, G_C) \mid H_1]$ .

SOLUTION. (i)

| $G_{AF} = g_1$ | $G_C = g_2$ | $\mathbb{P}(G_{AF}, G_C \mid H_1)$                       |
|----------------|-------------|----------------------------------------------------------|
| $(a, a)$       | $(a, a)$    | $p_a^3$                                                  |
| $(a, a)$       | $(a, b)$    | $p_a^2 p_b$                                              |
| $(a, b)$       | $(a, a)$    | $2p_a p_b \cdot \frac{1}{2} \cdot p_a = p_a^2 p_b$       |
| $(a, b)$       | $(a, b)$    | $2p_a p_b \cdot \frac{1}{2} \cdot (p_a + p_b) = p_a p_b$ |
| $(a, b)$       | $(b, b)$    | $2p_a p_b \cdot \frac{1}{2} \cdot p_b = p_a p_b^2$       |
| $(b, b)$       | $(a, b)$    | $p_a p_b^2$                                              |
| $(b, b)$       | $(b, b)$    | $p_b^3$                                                  |

(ii) We first calculate the likelihood ratio for all cases.

| $G_{AF} = g_1$ | $G_C = g_2$ | $\mathbb{P}(G_{AF}, G_C \mid H_1)$ | $\mathbb{P}(G_{AF}, G_C \mid H_2)$ | $\text{LR}_{H_1, H_2}$ |
|----------------|-------------|------------------------------------|------------------------------------|------------------------|
| $(a, a)$       | $(a, a)$    | $p_a^3$                            | $p_a^4$                            | $\frac{1}{p_a}$        |
| $(a, a)$       | $(a, b)$    | $p_a^2 p_b$                        | $2p_a^3 p_b$                       | $\frac{1}{2p_a}$       |
| $(a, b)$       | $(a, a)$    | $p_a^2 p_b$                        | $2p_a^3 p_b$                       | $\frac{1}{2p_a}$       |
| $(a, b)$       | $(a, b)$    | $p_a p_b$                          | $4p_a^2 p_b^2$                     | $\frac{1}{4p_a p_b}$   |
| $(a, b)$       | $(b, b)$    | $p_a p_b^2$                        | $2p_a p_b^3$                       | $\frac{1}{2p_b}$       |
| $(b, b)$       | $(a, b)$    | $p_a p_b^2$                        | $2p_a p_b^3$                       | $\frac{1}{2p_b}$       |
| $(b, b)$       | $(b, b)$    | $p_b^3$                            | $p_b^4$                            | $\frac{1}{p_b}$        |

$$\mathbb{E}[\text{LR}_{H_1, H_2}(G_{AF}, G_C) \mid H_1] = \frac{p_a^3}{p_a} + \frac{2p_a^2 p_b}{2p_a} + \frac{p_a p_b}{4p_a p_b} + \frac{2p_a p_b^2}{2p_b} + \frac{p_b^3}{p_b} = \frac{5}{4}.$$

(iii)

| $G_{AF} = g_1$ | $G_C = g_2$ | $\mathbb{P}(G_{AF}, G_C \mid H_2)$ | $\mathbb{P}(G_{AF}, G_C \mid H_3)$ | $\text{LR}_{H_3, H_2}$ |
|----------------|-------------|------------------------------------|------------------------------------|------------------------|
| $(a, a)$       | $(a, a)$    | $p_a^4$                            | $p_a^2$                            | $\frac{1}{p_a^2}$      |
| $(a, a)$       | $(a, b)$    | $2p_a^3 p_b$                       | 0                                  | 0                      |
| $(a, b)$       | $(a, a)$    | $2p_a^3 p_b$                       | 0                                  | 0                      |
| $(a, b)$       | $(a, b)$    | $4p_a^2 p_b^2$                     | $2p_a p_b$                         | $\frac{1}{2p_a p_b}$   |
| $(a, b)$       | $(b, b)$    | $2p_a p_b^3$                       | 0                                  | 0                      |
| $(b, b)$       | $(a, b)$    | $2p_a p_b^3$                       | 0                                  | 0                      |
| $(b, b)$       | $(b, b)$    | $p_b^4$                            | $p_b^2$                            | $\frac{1}{p_b^2}$      |

$$\mathbb{E}[\text{LR}_{H_3, H_2}(G_{AF}, G_C) \mid H_1] = \frac{p_a^3}{p_a^2} + \frac{p_a p_b}{2p_a p_b} + \frac{p_b^3}{p_b^2} = \frac{3}{2}.$$

□

EXERCISE 2.3. Consider an alleged father  $AF$  with heterozygous genotype  $(a, b)$  for some alleles  $a \neq b$  on some locus with possible alleles  $a, b, c$ . Suppose that the population frequencies of these alleles are  $p_a = 0.2$ ,  $p_b = 0.1$ ,  $p_c = 0.7$ . Let  $C$  be another person with genotype  $G_C$  on that locus. We have hypotheses:

- $H_1$ :  $AF$  is the father of  $C$ .
- $H_2$ :  $AF$  is unrelated to  $C$ .

(i) Complete the table using numerical values.

| $g$      | $\mathbb{P}(G_C = g \mid AF = (a, b), H_1)$ | $\mathbb{P}(G_C = g \mid AF = (a, b), H_2)$ | LR |
|----------|---------------------------------------------|---------------------------------------------|----|
| $(a, a)$ |                                             |                                             |    |
| $(a, b)$ |                                             |                                             |    |
| $(b, b)$ |                                             |                                             |    |
| $(a, c)$ |                                             |                                             |    |
| $(b, c)$ |                                             |                                             |    |
| $(c, c)$ |                                             |                                             |    |

(ii) Compute  $\mathbb{E}[\text{LR} \mid H_1]$ .

(iii) Verify by a direct computation that  $\mathbb{E}[\text{LR} \mid H_2] = 1$ .

(iv) Compute  $\mathbb{E}[\text{LR}^{-1} \mid H_2]$ .

(v) Compute  $\mathbb{E}[\text{LR}^{-1} \mid H_1]$ .

(vi) Verify that

$$\mathbb{P}(\text{LR} \geq 2 \mid H_2) = \mathbb{P}(\text{LR} \geq 2 \mid H_1) \mathbb{E}[\text{LR}^{-1} \mid \text{LR} \geq 2, H_1]$$

by computing the three terms in this equation.

SOLUTION. (i)

| $g$      | $\mathbb{P}(G_C = g \mid AF = (a, b), H_1)$ | $\mathbb{P}(G_C = g \mid AF = (a, b), H_2)$ | LR   |
|----------|---------------------------------------------|---------------------------------------------|------|
| $(a, a)$ | $\frac{1}{2}p_a = 0.1$                      | $p_a^2 = 0.04$                              | 2.5  |
| $(a, b)$ | $\frac{1}{2}p_a + \frac{1}{2}p_b = 0.15$    | $2p_ap_b = 0.04$                            | 3.75 |
| $(b, b)$ | $\frac{1}{2}p_b = 0.05$                     | $p_b^2 = 0.01$                              | 5    |
| $(a, c)$ | $\frac{1}{2}p_c = 0.35$                     | $2p_ap_c = 0.28$                            | 1.25 |
| $(b, c)$ | $\frac{1}{2}p_c = 0.35$                     | $2p_bp_c = 0.14$                            | 2.5  |
| $(c, c)$ | 0                                           | $p_c^2 = 0.49$                              | 0    |

(ii)

$$\begin{aligned} \mathbb{E}[\text{LR} \mid H_1] &= \sum_g \text{LR}(g) \cdot \mathbb{P}(G_C = g \mid H_1) \\ &= 2.5 \times 0.1 + 3.75 \times 0.15 + 5 \times 0.05 + 1.25 \times 0.35 + 2.5 \times 0.35 + 0 = 2.375. \end{aligned}$$

(iii)

$$\begin{aligned} \mathbb{E}[\text{LR} \mid H_2] &= \sum_g \text{LR}(g) \cdot \mathbb{P}(G_C = g \mid H_2) \\ &= 2.5 \times 0.04 + 3.75 \times 0.04 + 5 \times 0.01 + 1.25 \times 0.28 + 2.5 \times 0.14 + 0 = 1. \end{aligned}$$

(iv) Since  $\text{LR}^{-1}(c, c) = \infty$ , we have  $\mathbb{E}[\text{LR}^{-1} \mid H_2] = \infty$ .

(v)

$$\begin{aligned} \mathbb{E}[\text{LR}^{-1} \mid H_1] &= \sum_{g: \mathbb{P}(G_C = g \mid H_1) > 0} \frac{\mathbb{P}(G_C = g \mid H_2)}{\mathbb{P}(G_C = g \mid H_1)} \mathbb{P}(G_C = g \mid H_1) \\ &= \sum_{g \neq (c, c)} \mathbb{P}(G_C = g \mid H_2) \\ &= 0.04 + 0.04 + 0.01 + 0.28 + 0.14 = 0.51. \end{aligned}$$

(vi)

$$\begin{aligned} \mathbb{P}(\text{LR} \geq 2 \mid H_1) &= 0.10 + 0.15 + 0.05 + 0.35 = 0.65, \\ \mathbb{P}(\text{LR} \geq 2 \mid H_2) &= 0.04 + 0.04 + 0.01 + 0.14 = 0.23, \end{aligned}$$

$$\mathbb{E}[\text{LR}^{-1} \mid \text{LR} \geq 2, H_1] = \frac{1}{\mathbb{P}(\text{LR} \geq 2 \mid H_1)} \sum_{g: \text{LR} \geq 2} \mathbb{P}(G_C = g \mid H_2) = \frac{23}{65}.$$

□

# The Uncertainty of the Likelihood Ratio

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## SECTION 3.1

## | The Nature of the Likelihood Ratio

Suppose that at the scene of a crime, a DNA profile  $\Gamma$  has been obtained which we may assume was left by the criminal  $C$ . There is a suspect  $S$  who has this particular DNA profile, and we write  $H_1$  for the hypothesis that  $S = C$  and, therefore, that  $S$  left the DNA trace. We write  $H_2$  for the hypothesis that the unknown  $C$  is unrelated to  $S$ . The population frequency of this profile will have a certain value  $\theta$ ; the likelihood ratio would then be  $\frac{1}{\theta}$ . We assume for simplicity that  $W$  has a density  $\chi_W$ . We write  $E_C$  and  $E_S$  for the events that  $C$  and  $S$ , respectively, have characteristic  $\Gamma$ .

Under  $H_1$  the DNA profile has been seen once (since we assume that  $S = C$ ) and the probability that this happens is

$$\int_0^1 \mathbb{P}(E \mid W = t) \chi_W(t) dt = \int_0^1 t \chi_W(t) dt = \mathbb{E}[W].$$

Under  $H_2$  we have seen the DNA profile twice, and the probability of this is given by

$$\int_0^1 t^2 \chi_W(t) dt = \mathbb{E}[W^2].$$

We conclude that the likelihood ratio is given by

$$\text{LR} = \frac{\mathbb{E}[W]}{\mathbb{E}[W^2]} = \frac{\mathbb{E}[W]}{(\mathbb{E}[W])^2 + \text{Var}(W)} = \frac{1}{\mathbb{E}[W] + \frac{\text{Var}(W)}{\mathbb{E}[W]}}. \quad (3.1)$$

The continuous version of *Bayes' theorem* gives

$$\chi_{W|E_C}(t) = \frac{\mathbb{P}(E_C \mid W = t) \chi_W(t)}{\mathbb{P}(E_C)} = \frac{\mathbb{P}(E_C \mid W = t) \chi_W(t)}{\int_0^1 \mathbb{P}(E_C \mid W = t) \chi_W(t) dt} = \frac{t \chi_W(t)}{\mathbb{E}[W]}.$$

Hence

$$\mathbb{E}[W \mid E_C] = \int_0^1 t \chi_{W|E_C}(t) dt = \int_0^1 \frac{t^2 \chi_W(t)}{\mathbb{E}[W]} dt = \frac{\mathbb{E}[W^2]}{\mathbb{E}[W]}.$$

Combining this with (3.1) we conclude that

$$\text{LR} = \frac{1}{\mathbb{E}[W \mid E_C]}.$$

EXAMPLE 3.1. The Beta distribution has density given by

$$\chi_W(t) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} t^{a-1} (1-t)^{b-1},$$

for all  $t \in (0, 1)$ , and where  $a, b > 0$  are parameters. Here  $\Gamma$  denotes the *Gamma function* defined as

$$\Gamma(z) := \int_0^\infty x^{z-1} e^{-x} dx,$$

for all  $z > 0$ . It has the property that  $\Gamma(1) = 1$  and  $\Gamma(z+1) = z\Gamma(z)$ , hence  $\Gamma(n) = (n-1)!$  for any integer  $n \geq 1$ . If  $W$  has such a Beta distribution, then we have

$$\mathbb{E}[W] = \frac{a}{a+b}.$$

It then follows from the continuous version of *Bayes' rule* that

$$\begin{aligned} \chi_{W|E_C}(t) &= \frac{t\chi_W(t)}{\mathbb{E}[W]} \\ &= t\chi_W(t) \cdot \frac{a+b}{a} \\ &= \frac{t(a+b)}{a} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} t^{a-1} (1-t)^{b-1} \\ &= \frac{\Gamma(a+b+1)}{\Gamma(a+1)\Gamma(b)} t^a (1-t)^{b-1} \\ &= \chi_{W_{a+1,b}} = \frac{a+1}{a+b+1}. \end{aligned}$$

It follows that the likelihood ratio is equal to  $\frac{a+b+1}{a+1}$ .

## SECTION 3.2

## | The Posterior Likelihood Ratio Distribution

Suppose that the model parameters that occur in the statistical model are summarized by  $\theta$  and that proponents of  $H_1$  and  $H_2$  each have their own distribution of  $\theta$ , denoted by  $f(\theta | H_1)$  and  $f(\theta | H_2)$  respectively. We can now write

$$\text{LR}(E) = \frac{\int \mathbb{P}(E | \theta, H_1) f(\theta | H_1) d\theta}{\int \mathbb{P}(E | \theta, H_2) f(\theta | H_2) d\theta}.$$

Assuming that the ratios below do not lead to  $0/0$ , let

$$\text{LR}_\theta(E) = \frac{\mathbb{P}(E | \theta, H_1)}{\mathbb{P}(E | \theta, H_2)}$$

be the likelihood ratio we have when the value of  $\theta$  is known, and let

$$\text{LR}(\theta) = \frac{f(\theta | H_1)}{f(\theta | H_2)}.$$

We can rewrite  $\text{LR}(E)$  as an integral of  $\text{LR}_\theta(E)$  against an appropriate measure. Indeed, we have (changing the integration variable in the denominator for clarity) that

$$\begin{aligned} \text{LR}(E) &= \frac{\int \mathbb{P}(E | \theta, H_1) f(\theta | H_1) d\theta}{\int \mathbb{P}(E | v, H_2) f(v | H_2) dv} \\ &= \int \frac{\mathbb{P}(E | \theta, H_1) f(\theta | H_1)}{\int \mathbb{P}(E | v, H_2) f(v | H_2) dv} \cdot \frac{\mathbb{P}(E | \theta, H_2)}{\mathbb{P}(E | \theta, H_2)} d\theta \\ &= \int \frac{\mathbb{P}(E | \theta, H_1)}{\mathbb{P}(E | \theta, H_2)} \cdot \frac{\mathbb{P}(E | \theta, H_2) f(\theta | H_1)}{\int \mathbb{P}(E | v, H_2) f(v | H_2) dv} d\theta \\ &= \int \frac{\mathbb{P}(E | \theta, H_1)}{\mathbb{P}(E | \theta, H_2)} \cdot \frac{f(\theta | H_1)}{f(\theta | H_2)} \cdot \frac{\mathbb{P}(E | \theta, H_2) f(\theta | H_1)}{\int \mathbb{P}(E | v, H_2) f(v | H_2) dv} d\theta \\ &= \int \text{LR}_\theta(E) \text{LR}(\theta) \cdot \frac{\mathbb{P}(E | \theta, H_2) f(\theta | H_2)}{\int \mathbb{P}(E | v, H_2) f(v | H_2) dv} d\theta = \int \text{LR}_\theta(E) \text{LR}(\theta) f(\theta | E, H_2) d\theta, \end{aligned}$$

where  $f(\theta | E, H_2)$  is the distribution for  $\theta$  supplied by  $H_2$ , updated with the occurrence of the evidence  $E$ , and where we used the continuous version of *Bayes' theorem* in the last identity. If there is no disagreement concerning the distribution of  $\theta$ , then  $f(\theta | H_1)$  and  $f(\theta | H_2)$  coincide, and the term  $\text{LR}(\theta)$  disappears. In that case we get

$$\text{LR}(E) = \int \text{LR}_\theta(E) f(\theta | E, H_2) d\theta. \quad (3.2)$$

EXAMPLE 3.2.

EXAMPLE 3.3 (DNA Mixtures). Consider a suspect  $S$  with profile  $g$ . We consider the hypotheses  $H_1$  that  $S$  contributed to  $M$  versus  $H_2$  that he did not. Let  $E$  denote the event that mixture  $M$  has been observed, together with the observation that  $S$  is not excluded as a donor. Note that  $E$  does not reveal the precise profile  $g$  of  $S$ , only that  $S$  is not excluded as a donor to the mixture  $M$ . It is easy to see that

$$\text{LR}(E) = \frac{1}{RMNE(M)},$$

where the probability that a random person has only alleles that have been observed is often denoted  $RMNE(M)$  (Random Man Not Excluded), and  $M$  denotes the mixture.

From (3.2) we see that it is true that (replacing the integral by a sum and the density by a probability, since  $g$  can take only finitely many values)

$$\sum_g LR_g(E) \mathbb{P}(g \mid E, H_2) = LR(E).$$

Furthermore, (3.10) in this case yields

$$\sum_g LR_g(E) \mathbb{P}(g \mid E, H_1) \geq LR(E).$$

# The Bayesian Framework in Legal Cases

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## SECTION 4.1

## | The Sally Clark Case

In December 1996, Sally Clark's firstborn son Christopher was found dead in his bed, aged  $2\frac{1}{2}$  months. He died when Sally was alone in the house with him. In January 1998, her second child, Harry, died, aged 2 months, in similar circumstances. Soon afterwards, Sally was accused of having murdered her two children by smothering.

Apart from the possibility of a double murder, the possibility of a double Sudden Infant Death Syndrome (SIDS) was considered. The principal prosecution evidence came from a paediatrician, Sir Roy Meadow, who testified that the expected incidence of a child dying from a Sudden Infant Death Syndrome was about 1 in 8500. He also testified that a second occurrence of SIDS given a first one would be as likely as a first one, resulting in an overall rate of a double SIDS of about 1 in 73 million. This number, apparently, was so overwhelmingly small that a double SIDS was no longer deemed realistic, and since the only possible alternative that was considered was a double murder, this is what was concluded. Sally Clark was convicted of double murder.

We consider

- $H_1$ : Sally Clark murdered her two children by smothering.
- $H_2$ : The two children died of SIDS.
- $E$ : The two children died.

We can invoke *Bayes' rule*:

$$\frac{\mathbb{P}(H_1 | E)}{\mathbb{P}(H_2 | E)} = \frac{\mathbb{P}(E | H_1)}{\mathbb{P}(E | H_2)} \times \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_2)} = \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_2)} = \frac{\mu}{\delta},$$

where we write  $\mu$  for the prior probability of a double smothering, and  $\delta$  for the prior probability of a double SIDS. If we assume that the circumstances of the two dead children were such that the only possible explanations are double smothering or double SIDS, then we compute

$$\mathbb{P}(E | H_1^c) = \underbrace{\mathbb{P}(E | H_1^c, H_2)}_{=1} \mathbb{P}(H_2 | H_1^c) + \underbrace{\mathbb{P}(E | H_1^c, H_2^c)}_{=0} \mathbb{P}(H_2^c | H_1^c)$$

$$= \mathbb{P}(H_2 | H_1^c) = \frac{\mathbb{P}(H_2)}{\mathbb{P}(H_1^c)} = \frac{\delta}{1 - \mu}.$$

Hence the likelihood ratio of  $H_1$  versus  $H_1^c$  is  $\frac{1-\mu}{\delta}$  and the posterior odds are equal to

$$\frac{\mathbb{P}(H_1 | E)}{\mathbb{P}(H_1^c | E)} = \frac{1 - \mu}{\delta} \cdot \frac{\mu}{1 - \mu} = \frac{\mu}{\delta}.$$

These posterior odds are exactly the same as the posterior odds of  $H_1$  versus  $H_2$  that we discussed above.

## | Exercises

EXERCISE 4.1. In December 1996 Sally Clark's firstborn son Christopher was found dead in his bed, aged  $2\frac{1}{2}$  months. He died when Sally was alone in the house with him. In January 1998 her second child Harry also died, aged 2 months, in similar circumstances. Soon afterwards Sally Clark was accused of having murdered her two children by smothering. Apart from the possibility of a double murder, the possibility of a double Sudden Infant Death Syndrome (SIDS) was considered, and the following hypotheses were formulated.

- $H_1$ : Sally Clark murdered her two children.
- $H_2$ : The two children died of SIDS.

(i) Denoting the evidence of two dead children by  $E$ , explain why

$$\text{LR}_{H_1, H_2}(E) = 1.$$

Now let  $H_1^c$  be the hypothesis that Sally Clark did not murder her two children. We write  $\mu$  for the prior probability of a double murder, and  $\delta$  for the prior probability of a double SIDS. We assume that the circumstances of the two dead children were such that the only possible explanations are double murder or double SIDS.

(ii) Show that

$$\text{LR}_{H_1, H_1^c}(E) = \frac{1 - \mu}{\delta}.$$

(iii) The likelihood ratio in (ii) is much larger than the one in (i). It therefore seems to matter which pair of hypotheses is used in court. Comment on this.

# DNA

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## SECTION 5.1

## | Population Frequencies Versus Profile Probabilities

We denote by  $G_C = g$  the event that  $C$  has DNA profile  $g$ , and by  $G_S = g$  the event that  $S$  has DNA profile  $g$ . To compute the likelihood ratio in favor of  $H_1 : S = C$  versus  $H_2 : S \neq C$ , conditional on  $G_C = g$ , we note that if  $S = C$  and  $G_C = g$ , then the probability that  $G_S = g$  is equal to 1. Hence we obtain

$$\begin{aligned} \text{LR}_{H_1, H_2}(G_S = g) &= \frac{\mathbb{P}(G_C = g, G_S = g \mid H_1)}{\mathbb{P}(G_C = g, G_S = g \mid H_2)} \\ &= \frac{\cancel{\mathbb{P}(G_S = g \mid H_1)}}{\cancel{\mathbb{P}(G_S = g \mid H_2)}} \cdot \frac{\mathbb{P}(G_C = g \mid G_S = g, H_1)}{\mathbb{P}(G_C = g \mid G_S = g, H_2)} \\ &= \frac{1}{\mathbb{P}(G_C = g \mid G_S = g, H_2)} \approx \frac{1}{\mathbb{P}(G_C = g)}. \end{aligned}$$

We compute profile probabilities in a very large or even infinite population. In such a population, the probability that the parents of a person are related is zero, and the alleles that a person inherits from the paternal and the maternal side are obtained as two independent random draws from the population of alleles that the parental generation has. In the generation of offspring, the allele proportions will not have changed, since for allele  $a$  we get proportion

$$q_a = \mathbb{P}_{a,a} + \frac{1}{2} \sum_{b \neq a} \mathbb{P}_{a,b} = p_a^2 + \sum_{b \neq a} p_a p_b = p_a^2 + p_a(1 - p_a) = p_a.$$

This phenomenon is called *Hardy-Weinberg equilibrium* (HWE).

We consider two loci that are on different chromosomes. Since the segregation of different chromosomes takes place independently of each other, the inheritance processes on the two loci also evolve independently.

However, if two loci are on the same chromosome, then we need to know more about these loci since then they do not evolve independently. For a person with haplotypes  $a_1/b_1$  and  $a_2/b_2$ , we can infer from biology that there is a number  $0 \leq \rho \leq \frac{1}{2}$ , such that an offspring of that person inherits any of the two haplotypes  $a_i/b_i$  with probability  $\frac{1-\rho}{2}$ , and any of the two haplotypes  $a_i/b_j$  ( $j \neq i$ ) with probability  $\frac{\rho}{2}$ . The haplotypes  $a_1/b_2$  and  $a_2/b_1$  are called *recombinant products*, and  $\rho$  is called the probability of *recombination*.

## SECTION 5.2

# Uncertainty About Allele Frequencies: The Dirichlet Distribution

The probability that two specific alleles are both of type  $a$  is equal to

$$\int_0^1 x^2 f_a(x) dx,$$

and for any random variable  $X$  we have that  $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ .

Suppose that in the population, the alleles on the locus under consideration can be of type  $a_1, \dots, a_n$ . Let  $X_i$  denote the random variable describing our knowledge about the population frequency of the allele of type  $a_i$ . The random vector  $(X_1, \dots, X_n)$  takes values in the set  $\{(x_1, \dots, x_n) : 0 \leq x_i \leq 1, \sum_{i=1}^n x_i = 1\}$ .

Since the frequencies of the  $n$  alleles sum up to 1, we only need a probability density for the first  $n - 1$  alleles; the frequency of the last one is then determined by the requirement that the sum is equal to 1. Hence we only need a joint distribution of  $(X_1, \dots, X_{n-1})$ . For notational convenience, we often write  $\alpha := (\alpha_1, \dots, \alpha_n)$  and

$$B(\alpha) := \frac{\prod_{i=1}^n \Gamma(\alpha_i)}{\Gamma(\sum_{i=1}^n \alpha_i)},$$

then the  $(n - 1)$ -dimensional Dirichlet distribution with positive parameter  $\alpha$  is defined via the density

$$f_\alpha(x_1, \dots, x_{n-1}) := \frac{1}{B(\alpha)} \prod_{i=1}^n x_i^{\alpha_i - 1},$$

where

$$x_n := 1 - \sum_{i=1}^{n-1} x_i.$$

Then

$$\mathbb{E}[X_1] = \frac{\alpha_1}{\alpha_0}, \quad \mathbb{E}[X_1^2] = \frac{(\alpha_1 + 1)\alpha_1}{(\alpha_0 + 1)\alpha_0}, \quad \mathbb{E}[X_1 X_2] = \frac{\alpha_1 \alpha_2}{(\alpha_0 + 1)\alpha_0},$$

where  $\alpha_0 = \sum_{i=1}^n \alpha_i$ .

Suppose we take a sample of  $N$  alleles, and the numbers of alleles of type  $a_1, \dots, a_n$  in the sample are denoted by the random variables  $Y_1, \dots, Y_n$ , respectively, where  $Y_n$  is redundant and can be written as  $Y_n = N - \sum_{i=1}^{n-1} Y_i$ . Using the *continuous Bayes' rule* we have

$$\begin{aligned} f_\alpha(x_1, \dots, x_{n-1} \mid Y_1 = y_1, \dots, Y_n = y_n) &\stackrel{\mathcal{D}}{=} \text{Dir}(\alpha_1 + y_1, \dots, \alpha_n + y_n) \\ &= \mathbb{P}(Y_1 = y_1, \dots, Y_n = y_n \mid X_1 = x_1, \dots, X_{n-1} = x_{n-1}) \frac{f_\alpha(x_1, \dots, x_{n-1})}{\mathbb{P}(Y_1 = y_1, \dots, Y_n = y_n)} \\ &= \frac{N!}{\prod_{i=1}^n y_i!} \prod_{i=1}^n x_i^{y_i} \cdot \frac{\frac{1}{B(\alpha)} \prod_{i=1}^n x_i^{\alpha_i - 1}}{\mathbb{P}(Y_1 = y_1, \dots, Y_n = y_n)} \\ &= \frac{1}{B(\alpha)} \cdot \frac{N!}{\prod_{i=1}^n y_i!} \cdot \frac{1}{\mathbb{P}(Y_1 = y_1, \dots, Y_n = y_n)} \prod_{i=1}^n x_i^{\alpha_i + y_i - 1} = \frac{1}{B(\alpha + y)} \prod_{i=1}^n x_i^{\alpha_i + y_i - 1}, \end{aligned}$$

that is, the conditional distribution of the allele frequencies given the sample is again Dirichlet with parameters  $\alpha_i + y_i$ . This property is called the *conjugacy* of the multinomial and the Dirichlet distributions.

## SECTION 5.3

## | Match Probabilities

We start by modelling allele frequencies with a random vector having the Dirichlet distribution  $\text{Dir}(\alpha_1, \dots, \alpha_n)$ . After conditioning on the obtained population sample with counts  $(y_1, \dots, y_n)$ , we obtain the Dirichlet distribution  $\text{Dir}(\alpha_1 + y_1, \dots, \alpha_n + y_n)$  for the allele frequencies. Forensic laboratories often model the allele frequencies using a random vector with a differently parametrised Dirichlet distribution, namely

$$\text{Dir}\left(\frac{1-\theta}{\theta}p_1, \dots, \frac{1-\theta}{\theta}p_n\right),$$

where  $p_i = \frac{y_i}{N}$  is the observed frequency of the allele of type  $i$  in the sample, and where  $0 \leq \theta \leq 1$  is a population-genetic interpretation. We can interpret this in our context of uncertainty about the allele frequencies if we can write

$$\alpha_i + y_i = \frac{1-\theta}{\theta}p_i,$$

where we can only view the parameters  $\frac{1-\theta}{\theta}p_i$  as the posterior parameters of a Dirichlet distribution that was updated with the population database if  $\theta < \frac{1}{N+1}$ . However, forensic laboratories typically use a larger value for  $\theta$ , in the range of up to 0.03.

Suppose that the genotype is  $(a_1, a_1)$ . Then the density  $\chi_C(t)$  of  $W \mid E_C$  is the density of the random variable  $X_1^2$  derived from the Dirichlet distribution with parameters

$$\left(\frac{1-\theta}{\theta}p_1 + 2, \frac{1-\theta}{\theta}p_2, \dots, \frac{1-\theta}{\theta}p_n\right).$$

The expectation of this random variable is equal to

$$\frac{\frac{1-\theta}{\theta}p_1 + 3}{\frac{1-\theta}{\theta} + 3} \cdot \frac{\frac{1-\theta}{\theta}p_1 + 2}{\frac{1-\theta}{\theta} + 2} = \frac{3\theta + (1-\theta)p_1}{1 + 2\theta} \cdot \frac{2\theta + (1-\theta)p_1}{1 + \theta}. \quad (5.1)$$

In the case where the characteristic  $\Gamma$  corresponds to being heterozygous with alleles  $a_1$  and  $a_2$ , we need the expectation of  $2X_1X_2$ , where the  $X_i$  follow a Dirichlet distribution with parameters

$$\left(\frac{1-\theta}{\theta}p_1 + 1, \frac{1-\theta}{\theta}p_2 + 1, \frac{1-\theta}{\theta}p_3, \dots, \frac{1-\theta}{\theta}p_n\right).$$

This expectation is equal to

$$2 \cdot \frac{\theta + (1-\theta)p_1}{1 + 2\theta} \cdot \frac{\theta + (1-\theta)p_2}{1 + \theta}. \quad (5.2)$$

These two formulas (5.1) and (5.2) are called the *match probabilities* and are used by virtually all forensic laboratories.

## SECTION 5.4

| Population Genetics and the  $\theta$ -Correction

Consider a randomly mating subpopulation, partly isolated from a larger population. We denote by  $N$  the subpopulation size, that is, the number of alleles under consideration. Upon tracing back the history of an allele, several things may happen. The allele may have appeared in the population by *immigration* from a different subpopulation, or the allele came into the population by means of a *mutation* of another allele. When we consider two arbitrary alleles in the population, we can also consider the possibility that when we trace back their respective lineages, these lineages *coalesce* before an immigration or mutation event occurs with either of them, that is, the two alleles are two copies of the same ancestral allele.

We assume that when going back in time, coalescence of two lineages happens after an exponentially distributed waiting time with parameter  $\frac{1}{N}$ . We say that coalescence occurs at rate  $\frac{1}{N}$ . The waiting times (when going backwards in time) for pairs of alleles are independent of each other. At the same time, the waiting time until either an immigration or a mutation event occurs in a single lineage is assumed to have an exponential distribution with parameter  $\frac{\mu}{2N}$ . We assume that this waiting time does not depend on any coalescence event and that the waiting times are also independent across different lineages.

When we trace back the two lineages, it also follows from elementary properties of the exponential distribution that the probability that coalescence of two lineages takes place before any mutation or immigration event occurs with either of them is equal to

$$\theta := \frac{\frac{1}{N}}{\frac{1}{N} + \frac{2\mu}{2N}} = \frac{1}{1 + \mu}.$$

Let  $p_a$  be the probability that in a population in equilibrium, any given allele is of type  $a$ , so  $p_a$  is the population frequency of  $a$ . If we now take two alleles, then the probability that they are both of type  $a$  can be expressed in terms of  $\theta$  as

$$\theta p_a + (1 - \theta) p_a^2.$$

## The General Allele Sampling Formula in the IBD Interpretation

THEOREM 5.1.

*If we have already sampled  $r$  alleles of type  $a$  and  $s$  alleles of type  $b$ , the probability that the next allele that we sample is of type  $a$  is equal to*

$$\frac{r\theta + (1 - \theta)p_a}{1 + (r + s - 1)\theta}. \quad (5.3)$$

*Proof.* When  $r + s = 0$ , (5.3) reduces to  $p_a$  and the formula is correct. When  $r + s = 1$ , if one allele of type  $a$  has been sampled (that is,  $r = 1, s = 0$ ), the probability that the next sampled allele is of type  $a$  is  $\theta + (1 - \theta)p_a$ , and if the first sampled allele was not of type  $a$  (that is,  $r = 0, s = 1$ ) then the probability that the second allele is type  $a$  is  $(1 - \theta)p_a$ , in accordance with (5.3).  $\square$

## The $\theta$ -Correction in Forensic Practice

We model allele frequencies  $(X_a, X_b)$  as a random vector distributed as

$$\text{Dir}\left(\frac{1-\theta}{\theta}p_a, \frac{1-\theta}{\theta}p_b\right),$$

then if we have sampled  $r + s$  alleles where  $r$  were of type  $a$ , the distribution of the allele frequencies conditional on this sample is

$$\text{Dir}\left(\frac{1-\theta}{\theta}p_a + r, \frac{1-\theta}{\theta}p_b + s\right).$$

## SECTION 5.5

## | Kinship

## The Inbreeding Coefficient of an Individual

DEFINITION 5.2.

Individuals  $X$  and  $Y$  are said to be *related* if they have at least one ancestor in common. If the parents of a person are related, we say that this person is *inbred*.

If two persons are related, there exists a connected pedigree  $\mathcal{P}$  that contains both these persons, with at least one common ancestor. There is a non-zero probability that such a common ancestor has passed on the same allele via the pedigree to both individuals.

We may wonder how a pedigree affects their genetic similarity. One measure of this is the probability that a randomly selected allele from  $X$  is *IBD* (within the pedigree) to a randomly selected allele from  $Y$  (at the same locus, of course). This probability is called the *coancestry coefficient* or *kinship coefficient* and is denoted by  $\theta_{X,Y}$ .

For any individual  $X$  we denote by  $F_X$  the probability that the two alleles that  $X$  has on a given locus are *IBD*. For a child  $I$  of  $X$  and  $Y$ , we have  $F_I = \theta_{X,Y}$  since the two alleles of  $I$  (on a given locus) can be seen as random choices of the alleles of the parents  $X$  and  $Y$ .

EXAMPLE 5.3. (i) For an individual  $X$  we have  $\theta_{X,X} = \frac{1+F_X}{2}$ . Indeed, if we sample two random alleles from  $X$ , then with probability  $\frac{1}{2}$  we sample the same alleles, and then they are *IBD*. If we sample different alleles, then they are *IBD* with probability  $F_X$ .

(ii) Suppose that  $X$  is a parent of  $Y$  and that the parents of  $Y$  are unrelated. If the alleles of  $X$  are *IBD*, then two randomly chosen alleles from  $X$  and  $Y$  are certainly *IBD* if for  $Y$  we choose the allele that is inherited from  $X$ . This happens with probability  $\frac{F_X}{2}$ . If the alleles of  $X$  are not *IBD*, then the only way for the two alleles to be *IBD* is that the chosen allele at  $X$  is passed on to  $Y$ , and that this allele is chosen for  $Y$ . The probability of this is  $\frac{1-F_X}{4}$  so that

$$\theta_{X,Y} = \frac{F_X}{2} + \frac{1-F_X}{4} = \frac{1+F_X}{4}.$$

(iii) If  $X$  and  $Y$  are siblings with unrelated parents  $A$  and  $B$ , we claim that

$$\theta_{X,Y} = \frac{1}{4}\theta_{A,A} + \frac{1}{4}\theta_{B,B} = \frac{1}{4} + \frac{F_A + F_B}{8}.$$

To see this, observe that if we choose alleles at  $X$  and  $Y$  which come from different parents, then they cannot be *IBD* since we assume that the parents are unrelated. If both chosen alleles come from parent  $A$ , then they are *IBD* with probability  $\theta_{A,A} = \frac{1+F_A}{2}$ . The total probability of this possibility is, therefore, equal to  $\frac{1}{4} \times \frac{1+F_A}{2}$ . The computation in the case where both chosen alleles come from  $B$  is similar, leading to the claimed formula.

## Computing the Coancestry Coefficient for Non-Inbred Individuals

Consider two non-inbred individuals  $X$  and  $Y$  who are related to each other in a pedigree.<sup>1</sup> For each common ancestor  $A$ , there is a unique path in the pedigree from  $A$  to  $X$  and also a unique path from  $A$  to  $Y$ . Let there be  $n_1 \geq 0$  allele transmissions from  $A$  to  $X$  and  $n_2 \geq 0$  from  $A$  to  $Y$ .

If we select a random allele in  $X$ , the probability that this allele is IBD with an allele in  $A$  is equal to  $(\frac{1}{2})^{n_1}$ . Hence, if we take a random allele  $a$  in  $X$  and  $b$  in  $Y$ , the probability that both of them are equal to an allele in  $A$  is  $(\frac{1}{2})^{n_1+n_2}$ . The probability that these two alleles in  $A$  are furthermore IBD to each other is  $\theta_{A,A} = \frac{1+F_A}{2}$ . Thus, the probability that these alleles form a pair of IBD alleles originating from ancestor  $A$  is equal to  $(\frac{1}{2})^{n_1+n_2+1}(1+F_A)$ . Note that  $n_A = n_1 + n_2 + 1$  is the number of individuals in the path  $X \leftarrow A \rightarrow Y$  in the pedigree.

Of course,  $X$  and  $Y$  may have several common ancestors. However, when we consider a random allele  $a$  of  $X$  and a random allele  $b$  of  $Y$ , they can only be IBD if they are copies of an allele of one and the same common ancestor since we assume that the ancestors are unrelated. Thus, summing over all the common ancestors of  $X$  and  $Y$  now leads to the formula

$$\theta_{X,Y} = \sum_A \left(\frac{1}{2}\right)^{n_A} (1 + F_A).$$

## Likelihood Ratios for Pairwise Relatedness

Let  $X$  and  $Y$  be connected by a pedigree  $\mathcal{P}$  without inbreeding. We define  $\kappa_i(X, Y)$  as the probability that  $X$  and  $Y$  have exactly  $i$  pairs of IBD alleles inherited via pedigree  $\mathcal{P}$ . The  $\kappa_i$  are called the *IBD coefficients*. We have

$$\theta_{X,Y} = \frac{1}{4}\kappa_1(X, Y) + \frac{1}{2}\kappa_2(X, Y).$$

Denoting by  $I$  the number of IBD pairs between the genotypes of  $X$  and  $Y$ , we have

$$\mathbb{P}(G_Y = g_Y \mid G_X = g_X, H\mathcal{P}) = \sum_{i=0}^2 \kappa_i(X, Y) \mathbb{P}(G_Y = g_Y \mid G_X = g_X, I = i).$$

For individuals  $X, Y$  in a given pedigree  $\mathcal{P}$ , let  $\kappa_{i,j}(X, Y)$  be the probability that  $X$  and  $Y$  have  $i$  pairs of IBD alleles on the first and  $j$  pairs on the second locus. The coefficient  $\kappa_{i,j}(X, Y)$  is a function of the recombination probability  $\rho$ .

**EXAMPLE 5.4.** The IBD coefficients for half-siblings are  $\kappa_0 = \kappa_1 = \frac{1}{2}$  and  $\kappa_2 = 0$ . Two half-siblings have no IBD alleles if and only if they receive different alleles from their common parent. This is possible when the common parent passes on alleles without recombination twice (in which case the probability of there being no IBD alleles between  $X$  and  $Y$  is  $\frac{1}{2}$ ), or with recombination twice (with the same result), but not if there is recombination towards one offspring and not towards the other one. Thus,

$$\kappa_{0,0}(X, Y) = \frac{(1-\rho)^2}{2} + \frac{\rho^2}{2} = \frac{1}{2} - \rho + \rho^2.$$

$\kappa_{0,1}(X, Y) + \kappa_{0,0}(X, Y) = \kappa_0$  since the two possibilities on the left-hand side together give the probability that there are no IBD alleles on the first locus. Therefore,  $\kappa_{0,1}(X, Y) = \rho - \rho^2$ , and

<sup>1</sup> Their common ancestors are also their most recent common ancestors.

we similarly obtain for a grandparent-grandchild pair

$$\kappa_{1,0}(X, Y) = \kappa_{0,1}(X, Y) = \frac{\rho}{2}, \quad \kappa_{1,1}(X, Y) = \kappa_{0,0}(X, Y) = \frac{1-\rho}{2}.$$

Indeed, any child of the grandparent has one chromosome with alleles which come from the grandparent under consideration. The probability that none of these will be in the grandchild's genotype is  $\frac{1-\rho}{2}$ , which is therefore precisely equal to  $\kappa_{0,0}$ .

### Expectation of the Likelihood Ratio for Pairwise Kinship

If we consider genotypes  $(g_X, g_Y)$  from persons with IBD coefficients  $\kappa_i(X, Y)$ , then we write the probability to obtain them as

$$\mathbb{P}(g_X, g_Y \mid H_\kappa) = \sum_{i=0}^2 \kappa_i(X, Y) \mathbb{P}(g_X, g_Y \mid H_i), \quad (5.4)$$

where  $H_i$  is the hypothesis that  $\kappa_i(X, Y) = 1$ . We also let  $\text{LR}_i$  for  $i \in \{0, 1, 2\}$  be the likelihood ratio of  $H_i$  versus  $H_0$ , and  $\text{LR}_\kappa$  be the likelihood ratio that compares the hypotheses of being related according to  $\kappa$ , versus being unrelated. Then, since obviously  $\text{LR}_0 = 1$ , we have

$$\text{LR}_\kappa(g_X, g_Y) = \kappa_0(X, Y) + \kappa_1(X, Y)\text{LR}_1(g_X, g_Y) + \kappa_2(X, Y)\text{LR}_2(g_X, g_Y). \quad (5.5)$$

From (5.4) and (5.5) it now follows that we can compute the expectation of this random variable as

$$\mathbb{E}[\text{LR}_\kappa \mid H_{\kappa'}] = \sum_{i,j=0}^2 \kappa_i \kappa'_j \mathbb{E}[\text{LR}_i \mid H_j].$$

## Exercises

EXERCISE 5.1. Consider the pedigree in [Figure 5.1](#).

- (i) Calculate the IBD coefficients between persons 4 and 8, assuming that the founders of the pedigree are possibly inbred.
- (ii) Suppose that individual 4 has genotype  $(a, a)$ . What is, expressed in the inbreeding coefficients and the allele frequencies, the probability that individual 8 has genotype  $(a, b)$ ?
- (iii) Now assume all founders are outbred.
  - $H_1$ : Individuals 4 and 8 are related as in the pedigree of [Figure 5.1](#).
  - $H_2$ : Individuals 4 and 8 are an outbred half-sibling pair (and have no other mutual genetic relationship).

If individuals 4 and 8 both have genotype  $(a, a)$ , calculate the likelihood ratio.

- (iv) When person 1 is outbred and person 2 is fully inbred, the IBD coefficients for persons 4 and 8 are  $(\kappa_0, \kappa_1, \kappa_2) = (\frac{3}{8}, \frac{1}{2}, \frac{1}{8})$ . Give a pedigree with outbred founders in which there are two persons with the same IBD coefficients.

SOLUTION. (i) The transmissions from founder 1 (to 4 and to 8) are  $\text{IBD}$  with probability

$$\mathbb{P}_1 = F_1 + \frac{1 - F_1}{2} = \frac{1 + F_1}{2}.$$

Two transmissions from founder 2 (to 4 and to 5) are  $\text{IBD}$  with probability  $\frac{1+F_2}{2}$ . Then, the allele from 5 must be passed down to 8 through  $5 \rightarrow 7 \rightarrow 8$ , which happens with probability  $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ . Hence

$$\mathbb{P}_2 = \frac{1 + F_2}{2} \cdot \frac{1}{4} = \frac{1 + F_2}{8}.$$

Therefore

$$\begin{aligned}\kappa_2 &= \mathbb{P}_1 \mathbb{P}_2 = \frac{(1 + F_1)(1 + F_2)}{16}, \\ \kappa_0 &= (1 - \mathbb{P}_1)(1 - \mathbb{P}_2) = \frac{(1 - F_1)(7 - F_2)}{16}, \\ \kappa_1 &= 1 - \kappa_0 - \kappa_2 = \frac{4 + 3F_1 - F_1 F_2}{8}.\end{aligned}$$

(ii) If  $I = 2$ ,  $G_8 = (a, a)$ , so the probability of  $G_8 = (a, b)$  is 0. Then

$$\mathbb{P}(G_8 = (a, b) \mid G_4 = (a, a)) = \kappa_0 \cdot 2p_a p_b + \kappa_1 \cdot p_b = p_b \left( \frac{(1 - F_1)(7 - F_2)}{8} p_a + \frac{4 + 3F_1 - F_1 F_2}{8} \right).$$

(iii) From (i),  $(\kappa_0, \kappa_1, \kappa_2)_{H_1} = (\frac{7}{16}, \frac{1}{2}, \frac{1}{16})$ . For outbred half-siblings,  $(\kappa'_0, \kappa'_1, \kappa'_2)_{H_2} = (\frac{1}{2}, \frac{1}{2}, 0)$ . Therefore the likelihood ratio is

$$\text{LR} = \frac{\mathbb{P}(G_8 = (a, a) \mid G_4 = (a, a), H_1)}{\mathbb{P}(G_8 = (a, a) \mid G_4 = (a, a), H_2)} = \frac{\kappa_0 p_a^2 + \kappa_1 p_a + \kappa_2}{\kappa'_0 p_a^2 + \kappa'_1 p_a + \kappa'_2} = \frac{7p_a^2 + 8p_a + 1}{8p_a(1 + p_a)}.$$

(iv) A woman  $M$  has two children  $X$  and  $Y$  with two different men  $F_1$  and  $F_2$ . Half-siblings give an  $\text{IBD}$  pair from the maternal side with probability  $\frac{1}{2}$ . Make  $F_1$  and  $F_2$  full brothers (same outbred parents  $A$  and  $B$ ).  $X$  and  $Y$  are cousins, giving an  $\text{IBD}$  pair from the paternal side with probability  $\frac{1}{4}$ .

□

EXERCISE 5.2. In the population-genetics interpretation, the parameter  $\theta$  is interpreted as the probability for two alleles to be  $\text{IBD}$ . Suppose for a given locus only two alleles  $a$  and  $b$  are possible. If one has already sampled  $r$  alleles of type  $a$  and  $s$  alleles of type  $b$ , then the probability that the next allele we sample is of type  $a$  is equal to

$$\frac{r\theta + (1 - \theta)p_a}{1 + (r + s - 1)\theta}, \quad (5.6)$$

where  $p_a$  is the population frequency of  $a$ .

(i) Show that this is also the formula one obtains starting from a Dirichlet distribution of the allele frequencies with parameters  $(\frac{1-\theta}{\theta}p_a, \frac{1-\theta}{\theta}p_b)$ .

(ii) Now consider three limits in (5.6):

- $s$  constant,  $r \rightarrow \infty$ .
- $r$  constant,  $s \rightarrow \infty$ .
- $r \rightarrow \infty, s \rightarrow \infty, \frac{r}{s} \rightarrow \alpha \in (0, \infty)$ .

Compute the limit in each of these cases, and explain the outcomes.

(iii) There is one value of  $p_a$  for which the expression in (5.6) does not depend on  $\theta$ . Which value is that?

- (iv) Is the limit for  $\theta \rightarrow 1$  or  $\theta \rightarrow 0$  meaningful? If not, explain why; if yes explain the numerical value of the limit.

SOLUTION. (i) Let  $(X_a, X_b) \sim \text{Dir}(\alpha_a, \alpha_b)$ , where

$$\alpha_a = \frac{1-\theta}{\theta} p_a, \quad \alpha_b = \frac{1-\theta}{\theta} p_b.$$

After observing  $r$  alleles of type  $a$  and  $s$  alleles of type  $b$ , conjugacy gives  $\text{Dir}(\alpha_a + r, \alpha_b + s)$ . Then

$$\mathbb{E}[X_a \mid r, s] = \frac{\alpha_a + r}{\alpha_a + \alpha_b + r + s} = \frac{r + \frac{1-\theta}{\theta} p_a}{r + s + \frac{1-\theta}{\theta}} = \frac{r\theta + (1-\theta)p_a}{1 + (r+s-1)\theta}.$$

- (iii) If the expression is independent of  $\theta$ , it must take the same value at extremes. Hence we need

$$p_a = \frac{r}{r+s}.$$

- (iv)  $\theta \rightarrow 0$  tends to no  $\text{IBD}$  dependence (or very strong mutation/immigration), so the next allele behaves like an independent draw from the population.

$\theta \rightarrow 1$  means alleles are *a.s.*  $\text{IBD}$ , the dependence becomes very strong and the probability is driven by the observed sample composition.

□

EXERCISE 5.3. Consider the pedigree in [Figure 5.3](#).

- (i) Identify the inbred individuals, if any.
- (ii) Compute  $\theta_{X,Y}$  for individuals  $X = 5, Y = 6$  in this pedigree.
- (iii) Consider a locus with allele frequencies  $p_a = 0.2, p_b = 0.8$ . Suppose that individual 5 has genotype  $(a, a)$  on this locus. Give the probability distribution for the genotype of individual 6 on this locus.

SOLUTION. (i) No inbred individuals.

- (ii)  $\theta_{5,6} = \frac{1+F_5}{4} = \frac{1}{4}$ .

- (iii)  $\mathbb{P}(G_6 = (a, a) \mid G_5 = (a, a)) = 0.2$ ,  $\mathbb{P}(G_6 = (a, b) \mid G_5 = (a, a)) = 0.8$ , and  $\mathbb{P}(G_6 = (b, b) \mid G_5 = (a, a)) = 0$ .

□

EXERCISE 5.4. Consider the pedigree in [Figure 5.4](#).

- (i) Assume that all founders of the pedigree are outbred, i.e., have inbreeding coefficient equal to zero. Derive the  $\text{IBD}$  probabilities  $(\kappa_0, \kappa_1, \kappa_2)$  for individuals 10 and 14.
- (ii) Suppose the genotypes of individuals 10 and 14 are such that they have no alleles in common. Calculate the likelihood ratio.
  - $H_1$ : Individuals 10 and 14 are related as in the pedigree of [Figure 5.4](#).
  - $H_2$ : Individuals 10 and 14 are full siblings (whose parents are outbred).
- (iii) Suppose that we do allow for positive inbreeding coefficients in the founders of the pedigree in [Figure 5.4](#). Argue in which direction the likelihood ratio in (ii) will change irrespective of allele frequencies (i.e., increase or decrease), or argue that it may increase or decrease depending on the allele frequencies.

SOLUTION. (i) Individual 10's father 9 and individual 14's mother 13 are full siblings (offspring of founders 11 and 12), with  $\text{IBD}$  probability  $\theta_{9,13} = \frac{1}{4}$ .

Individual 10's mother, 6, and individual 14's father, 8, are related through founders 1 and 2, and

$$\theta_{6,8} = \frac{1}{2}\theta_{3,8} = \frac{1}{4}\theta_{3,4} = \frac{1}{16}.$$

Then

$$\kappa_2 = \theta_{6,8}\theta_{9,13} = \frac{1}{16} \cdot \frac{1}{4} = \frac{1}{64},$$

$$\kappa_0 = (1 - \theta_{6,8})(1 - \theta_{9,13}) = \left(1 - \frac{1}{16}\right)\left(1 - \frac{1}{4}\right) = \frac{45}{64},$$

$$\kappa_1 = 1 - \kappa_0 - \kappa_2 = \frac{9}{32}.$$

(ii) For outbred full siblings  $(\kappa'_0, \kappa'_1, \kappa'_2) = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ . Then

$$\text{LR} = \frac{\kappa_0(H_1)}{\kappa_0(H_2)} = 2.8125.$$

(iii) Positive founder inbreeding increases  $\text{IBD}$  probabilities, thus reducing  $\kappa_0(H_1)$ , then the likelihood ratio must decrease. □

EXERCISE 5.5. Consider the pedigree depicted in [Figure 5.5](#).

- (i) Identify the inbred individuals in this pedigree.
- (ii) Compute  $\theta_{X,Y}$ , that is, the probability that an allele chosen randomly from  $X$  is  $\text{IBD}$  to an allele randomly chosen from  $Y$ .
- (iii) Determine the probabilities  $\kappa_0, \kappa_1, \kappa_2$  for  $X$  and  $Y$  to share 0, 1, 2 pairs of alleles  $\text{IBD}$ .

SOLUTION. (i) An individual is inbred in a pedigree if their parents are related in that pedigree. Thus, persons 7 and 8 are inbred; the others are not.

(ii) If  $X$  is not an ancestor of  $Y$  (in particular,  $X \neq Y$ ), then

$$\begin{aligned} \theta_{X,Y} &= \frac{1}{2}\theta_{P_8,Y} + \frac{1}{2}\theta_{P_7,Y} \\ &= \frac{1}{2}\left(\frac{1}{2}\theta_{P_7,P_8} + \frac{1}{2}\theta_{P_8,P_8}\right) + \frac{1}{2}\left(\frac{1}{2}\theta_{P_7,P_8} + \frac{1}{2}\theta_{P_7,P_7}\right) \\ &= \frac{1}{4}\theta_{P_7,P_7} + \frac{1}{4}\theta_{P_8,P_8} + \frac{1}{2}\underbrace{\theta_{P_7,P_8}}_{=0} = \frac{1}{4}(\theta_{P_7,P_7} + \theta_{P_8,P_8}). \end{aligned}$$

If  $P_1$  has two alleles  $(a, b)$  that are not  $\text{IBD}$ , and suppose  $a$  is passed on from  $P_1$  to  $P_7$ , then the probability is  $\frac{1}{4}$  that this allele is also transmitted to  $P_7$  via  $P_1 \rightarrow P_5 \rightarrow P_7$ . Thus, with probability  $\frac{1}{4}$  the two alleles of  $P_7$  are  $\text{IBD}$ . If  $P_1$  has two  $\text{IBD}$  alleles  $(a, a)$ , then  $P_7$  receives as paternal allele  $a$ , and the probability is  $\frac{1}{2}$  that allele  $a$  is also transmitted to  $P_7$  via  $P_1 \rightarrow P_5 \rightarrow P_7$ . Hence,

$$F_7 = \frac{1}{4}(1 - F_1) + \frac{1}{2}F_1 = \frac{1 + F_1}{4}.$$

Then

$$\theta_{P_7,P_7} = \frac{1 + F_7}{2} = \frac{5 + F_1}{8}.$$

Similarly,  $\theta_{P_8,P_8} = \frac{5 + F_3}{8}$ . Hence,

$$\theta_{X,Y} = \frac{5}{16} + \frac{F_1 + F_3}{32}.$$

(iii)

$$\kappa_2(X, Y) = \theta_{P_7, P_7} \theta_{P_8, P_8} + \underbrace{\theta_{P_7, P_8} \theta_{P_8, P_7}}_{=0} = \frac{(5 + F_1)(5 + F_3)}{64}.$$

Since  $\theta_{X, Y} = \frac{1}{4}\kappa_1 + \frac{1}{2}\kappa_1$ ,

$$\begin{aligned}\kappa_1(X, Y) &= \frac{15 - F_1 - F_3 - F_1 F_3}{32}, \\ \kappa_0(X, Y) &= 1 - \kappa_1 - \kappa_2 = \frac{(3 - F_1)(3 - F_3)}{64}.\end{aligned}$$

□

EXERCISE 5.6. (i) Suppose that  $X$  and  $Y$  are maternal half-siblings with common mother  $M$  and unrelated fathers. Express  $\theta_{X, Y}$  in terms of the inbreeding coefficients of their parents.

(ii) Now suppose that it turns out that  $M$ 's parents are both children of the same anonymous sperm donor, i.e., are paternal half-siblings with unrelated mothers. Compute the inbreeding coefficient of  $M$  and use this to recompute  $\theta_{X, Y}$ . You may assume that the inbreeding coefficient of the sperm donor is zero.

(iii) Suppose that on some autosomal locus, there are 10 alleles in the population, each with population frequency 0.1. What is the probability that  $X$  and  $Y$  have the same genotype on this locus?

SOLUTION. (i)

$$\theta_{X, Y} = \frac{1}{4}\theta_{M, M} = \frac{1}{4} \cdot \frac{1 + F_M}{2} = \frac{1 + F_M}{8}.$$

(ii) Let the sperm donor be  $D$  with  $F_D = 0$ . Then

$$F_M = \frac{1}{4}\theta_{D, D} = \frac{1}{4} \cdot \frac{1 + F_D}{2} = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}.$$

So

$$\theta_{X, Y} = \frac{1 + F_M}{8} = \frac{1 + \frac{1}{8}}{8} = \frac{9}{64}.$$

(iii)

$$\kappa_1 = 4\theta_{X, Y} = 4 \cdot \frac{9}{64} = \frac{9}{16}, \quad \kappa_0 = 1 - \kappa_1 = \frac{7}{16}.$$

Therefore

$$\begin{aligned}\mathbb{P}(G_X = G_Y) &= \kappa_0 \left( \sum_i p_i^4 + \sum_{i < j} (2p_i p_j)^2 \right) + \kappa_1 \left( \sum_i p_i^2 \right) \\ &= \kappa_0 \left( 10p_i^4 + \binom{10}{2} 4p_i^2 p_j^2 \right) + \kappa_1 (10p_i^2) \\ &= \frac{7}{16} \times 0.019 + \frac{9}{16} \times 0.1 \approx 0.06456.\end{aligned}$$

□

EXERCISE 5.7. Consider the pedigree  $\mathcal{P}$  with individuals  $X$  and  $Y$  below.

(i) Compute the coancestry coefficient  $\theta_{X, Y}$ . The persons  $A_i$  in  $\mathcal{P}$  have inbreeding coefficients  $F_i \geq 0$ .

- (ii) If all  $F_i = 1$ , what are the IBD coefficients between  $S_1$  and  $S_2$ ? What standard relationship does that correspond to?
- (iii) If all  $F_i = 1$ , what are the IBD coefficients between  $B_1$  and  $B_2$ ? What standard relationship does that correspond to?
- (iv) Define a pedigree  $\mathcal{P}'$  with non-inbred founders such that  $X$  and  $Y$  have the same IBD coefficients as in the depicted pedigree  $\mathcal{P}$  where all  $F_i = 1$ .
- (v) Now let all  $F_i = 0$ . Derive the IBD coefficients for  $X$  and  $Y$  in  $\mathcal{P}$ .

SOLUTION. (i) There is no inbreeding in the pedigree, so we can use the path counting method. The paths from  $X$  to  $Y$  are  $X \rightarrow S_1 \rightarrow A_2 \rightarrow S_2 \rightarrow Y$ ,  $X \rightarrow B_2 \rightarrow A_4 \rightarrow B_1 \rightarrow Y$ ,  $X \rightarrow B_2 \rightarrow A_5 \rightarrow B_1 \rightarrow Y$ , so

$$\theta_{X,Y} = \frac{1 + F_2 + 1 + F_4 + 1 + F_5}{32}.$$

- (ii) In that case, the alleles that  $S_1$  and  $S_2$  obtain from  $A_2$  are IBD, and those of their other parent are not. Hence, they have  $(\kappa_0, \kappa_1, \kappa_2) = (0, 1, 0)$ . These are the coefficients that parents and children also have.
- (iii) In that case, the alleles that  $B_1$  and  $B_2$  obtain from  $A_4$  are IBD, and those of  $A_5$  too. Hence, they have  $(\kappa_0, \kappa_1, \kappa_2) = (0, 0, 1)$ . These are the coefficients that monozygotic twins also have.
- (iv) Replace  $S_1, S_2$  with a parent-child pair and  $B_1, B_2$  by monozygotic twins (or let  $B_1 = B_2$ ).
- (v)

$$\kappa_2 = \theta_{S_1, S_2} \theta_{B_1, B_2} + \underbrace{\theta_{S_1, B_2} \theta_{B_1, S_2}}_{=0} = \frac{1}{8} \times \frac{1}{4} = \frac{1}{32}.$$

Since  $\theta_{X,Y} = \frac{1}{4}\kappa_1 + \frac{1}{2}\kappa_2$  and  $\theta_{X,Y} = \frac{3}{32}$ , we get  $\kappa_1 = \frac{5}{16}$  and then  $\kappa_0 = \frac{21}{32}$ .

□

EXERCISE 5.8. Consider  $X$  and  $Y$  who are first cousins.

- (i) Compute  $\theta_{X,Y}$ , i.e., the probability that an allele that is randomly selected from  $X$  is IBD to an allele that is randomly selected from  $Y$ , assuming that the founders  $A$  and  $B$  have inbreeding coefficients  $F_A \geq 0$  and  $F_B \geq 0$  respectively.
- (ii) Suppose that  $A$  and  $B$  are first cousins. Express  $\theta_{X,Y}$  in terms of the inbreeding coefficients of the grandparents of  $A$  and  $B$ .

SOLUTION. (i)  $X$  and  $Y$  are not inbred since their parents are not related. Therefore we can use the path counting formula from the lectures. There are two paths of length 5 each, via  $A$  or  $B$ . Thus,

$$\theta_{X,Y} = \frac{1 + F_A + 1 + F_B}{32} = \frac{1}{16} + \frac{F_A + F_B}{32}.$$

- (ii) Now  $X$  and  $Y$  have become inbred since their parents are related now. Then

$$\begin{aligned} \theta_{X,Y} &= \frac{1}{2}\theta_{F(X),Y} + \underbrace{\frac{1}{2}\theta_{M(X),Y}}_{=0} \\ &= \frac{1}{2} \left( \frac{1}{2}\theta_{F(X),F(Y)} + \frac{1}{2}\underbrace{\theta_{F(X),M(Y)}}_{=0} \right) \\ &= \frac{1}{4} \left( \frac{1}{2}\theta_{A,F(Y)} + \frac{1}{2}\theta_{B,F(Y)} \right) \end{aligned}$$

$$= \frac{1}{8} \left( \frac{1}{2} \theta_{A,B} + \frac{1}{2} \theta_{A,A} + \frac{1}{2} \theta_{A,B} + \frac{1}{2} \theta_{B,B} \right) = \frac{1}{16} (\theta_{A,A} + \theta_{B,B} + 2\theta_{A,B}).$$

Indeed,  $A$  and  $B$  are not inbred; hence,  $\theta_{A,A} = \theta_{B,B} = \frac{1}{2}$ . Since  $A$  and  $B$  are first cousins, we can use part (i) to conclude that

$$\theta_{A,B} = \frac{1}{16} + \frac{F_C + F_D}{32},$$

where  $C$  and  $D$  are the grandparents that  $A$  and  $B$  have in common. Filling this in gives

$$\theta_{X,Y} = \frac{1}{16} + \frac{1}{8} \theta_{A,B} = \frac{9}{128} + \frac{F_C + F_D}{256}.$$

□

EXERCISE 5.9. Consider the pedigree in *Figure A25-2Q1*. Assume that the founders may be inbred.

- (i) Compute  $\theta_{X,Y}$ , assuming all founders may be inbred.
- (ii) Compute the IBD coefficients  $\kappa_i(X, Y)$ .
- (iii) Now let the founders be outbred. Suppose the genotypes of  $X$  and  $Y$  are  $g_X = (a, b)$  and  $g_Y = (a, b)$ . Express the LR for  $X, Y$  to be related as shown in *Figure A25-2Q1*, versus unrelated, in terms of the allele frequencies.

SOLUTION. (i)

$$\theta_{X,Y} = \left(\frac{1}{2}\right)^3 (1 + F_3) + \left(\frac{1}{2}\right)^4 (1 + F_2) = \frac{1 + F_3}{8} + \frac{1 + F_2}{16}.$$

(ii)

$$\begin{aligned} \kappa_2 &= \theta_{2,4} \theta_{3,3} = \frac{(1 + F_2)(1 + F_3)}{8}, \\ \kappa_1 &= 4\theta_{X,Y} - 2\kappa_2 = \frac{2 + F_3 - F_2 F_3}{4}, \\ \kappa_0 &= 1 - \kappa_1 - \kappa_2 = \frac{(3 - F_2)(1 - F_3)}{8}. \end{aligned}$$

(iii) Outbred founders imply that  $F_i = 0$ .

$$\begin{aligned} \text{LR} &= \sum_{i=0}^2 \kappa_i \frac{\mathbb{P}(g_X, g_Y \mid I = i)}{\mathbb{P}(g_X, g_Y \mid I = 0)} \\ &= \frac{3}{8} + \frac{1}{2} \cdot \frac{\mathbb{P}(g_X, g_Y \mid I = 1)}{\mathbb{P}(g_X, g_Y \mid I = 0)} + \frac{1}{8} \cdot \frac{\mathbb{P}(g_X, g_Y \mid I = 2)}{\mathbb{P}(g_X, g_Y \mid I = 0)} \\ &= \frac{3}{8} + \frac{1}{2} \cdot \frac{\frac{1}{2}(p_a + p_b)}{2p_a p_b} + \frac{1}{8} \cdot \frac{1}{2p_a p_b} = \frac{6p_a p_b + 2(p_a + p_b) + 1}{16p_a p_b}. \end{aligned}$$

□

EXERCISE 5.10. Consider the pedigree in *Figure A25-2Q2*. A woman  $X$  accuses her relative  $S$  of being the father of her child  $C$ .

- (i) If this pedigree is correct, what is the probability (prior to observing any genotypes) that the child is homozygous on an autosomal locus with allele frequencies  $p_a$ ? You may assume all founders of the pedigree are outbred.

The relative  $S$ , not yet formally a suspect, is not obliged to provide a DNA reference sample for a paternity test, and refuses. We therefore only consider genetic data of  $X$  and  $C$ . We set

up hypotheses  $H_1$  being the woman's claim versus  $H_2$  that the father of her child is a man unrelated to her.

- (ii) On an autosomal locus,  $g_X = (a, a)$  and  $g_C = (a, a)$  are the genotypes of mother and child. Determine the probabilities  $\mathbb{P}(g_S = g \mid g_X = (a, a))$  for all  $g$ .
- (iii) Determine, e.g., using the results in (ii), the likelihood ratio for  $H_1$  versus  $H_2$  for these data as a function of  $p_a$ .

SOLUTION. (i)

$$\begin{aligned} P_{\text{homozygous}} &= \sum_a (Fp_a + (1-F)p_a^2) \\ &= F \sum_a p_a + (1-F) \sum_a p_a^2 \\ &= F + (1-F) \sum_a p_a^2 = \frac{5}{32} + \frac{27}{32} \sum_a p_a^2, \end{aligned}$$

where we calculate that

$$F = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \theta_{4,6} = \frac{1}{8} + \frac{1}{4} \left( \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \right) = \frac{5}{32}.$$

(ii) Given that

$$P(5 = (a, a)) = p_a^2, \quad P(5 = (a, b)) = p_a p_b, \quad P(5 = (b, b)) = p_b^2, \quad (b \neq a)$$

we have

$$P(5 = (a, a) \mid X \leftarrow 5 = a) = p_a, \quad P(5 = (a, b) \mid X \leftarrow 5 = a) = p_b.$$

Then we calculate:

$$P(S \leftarrow 5 = a \mid X \leftarrow 5 = a) = p_a + \frac{1}{2} \sum_{b \neq a} p_b = p_a + \frac{1-p_a}{2} = \frac{1+p_a}{2}, \quad (\sum_{b \neq a} p_b = 1-p_a)$$

$$P(S \leftarrow 5 = b \mid X \leftarrow 5 = a) = \frac{p_b}{2},$$

$$P(S \leftarrow 4 = a \mid X \leftarrow 6 = a) = \theta_{4,6} + (1-\theta_{4,6})p_a = \frac{1+7p_a}{8}, \quad (\theta_{4,6} = \frac{1}{8} \text{ by part (i)})$$

$$P(S \leftarrow 4 = b \mid X \leftarrow 6 = a) = (1-\theta_{4,6})p_b = \frac{7p_b}{8}.$$

Therefore,

$$\begin{aligned} P(g_S = (a, a) \mid g_X = (a, a)) &= \frac{1+p_a}{2} \cdot \frac{1+7p_a}{8} = \frac{1+8p_a+7p_a^2}{16}, \\ P(g_S = (a, b) \mid g_X = (a, a)) &= \frac{1+p_a}{2} \cdot \frac{7p_b}{8} + \frac{p_b}{2} \cdot \frac{1+7p_a}{8} = \frac{4p_b+7p_a p_b}{8}, \\ P(g_S = (b, b) \mid g_X = (a, a)) &= \frac{p_b}{2} \cdot \frac{7p_b}{8} = \frac{7p_b^2}{16}. \end{aligned}$$

(iii)

$$\begin{aligned} LR_{H_1, H_2} &= \frac{P(G_X = (a, a), G_C = (a, a) \mid H_1)}{P(G_X = (a, a), G_C = (a, a) \mid H_2)} \\ &= \frac{\frac{1+8p_a+7p_a^2}{16} + \frac{1}{2} \sum_{b \neq a} \frac{4p_b+7p_a p_b}{8}}{p_a} \\ &= \frac{1+8p_a+7p_a^2 + \frac{4+7p_a}{16} \sum_{b \neq a} p_b}{16p_a} \\ &= \frac{1+8p_a+7p_a^2 + (4+7p_a)(1-p_a)}{16p_a} = \frac{5+11p_a}{16p_a}. \end{aligned}$$

□

EXERCISE 5.11. From a DNA trace, a single-source DNA profile on 10 loci is obtained. A match between the trace and a suspect, upon comparing these 10 loci, yields a match probability of 1 in 1 billion. The court wants to know the probability that, when additional profiling is performed on the trace to obtain 5 more loci that can be compared with the suspect, the profiles will still match. What do you answer the court?

SOLUTION. We define:

- $H_1$ : The suspect is the source of the DNA trace.
- $H_2$ : The suspect is not the source of the DNA trace.
- $E_1$ : The 10 loci match between the suspect and the trace.
- $E_2$ : The 5 more loci also match.

Let  $P(H_1) = \pi$ , so that  $P(H_2) = 1 - \pi$ . By *Bayes' Rule*, we have

$$\frac{P(H_1 | E_1)}{P(H_2 | E_1)} = \frac{P(E_1 | H_1)}{P(E_1 | H_2)} \cdot \frac{P(H_1)}{P(H_2)} = \frac{1}{10^{-9}} \cdot \frac{\pi}{1 - \pi} = \frac{10^9 \pi}{1 - \pi}.$$

Since  $P(H_1 | E_1) + P(H_2 | E_1) = 1$ , we obtain

$$P(H_1 | E_1) = \frac{10^9 \pi}{10^9 \pi + 1 - \pi}, \quad P(H_2 | E_1) = \frac{1 - \pi}{10^9 \pi + 1 - \pi}.$$

Therefore,

$$\begin{aligned} P(E_2 | E_1) &= \underbrace{P(E_2 | E_1, H_1)}_{=1} P(H_1 | E_1) + P(E_2 | E_1, H_2) P(H_2 | E_1) \\ &= P(H_1 | E_1) + P(E_2 | H_2) P(H_2 | E_1) \\ &= \frac{10^9 \pi}{10^9 \pi + 1 - \pi} + \frac{1 - \pi}{10^9 \pi + 1 - \pi} \underbrace{P(E_2 | H_2)}_{\text{very small}} \\ &\approx \frac{10^9 \pi}{10^9 \pi + 1 - \pi} = P(H_1 | E_1). \end{aligned}$$

In conclusion, we should explain to the court that the probability of 5 more loci matching depends on the posterior probability that the suspect is the source after the current DNA evidence is taken into account. This posterior probability is for the court to assess. □

EXERCISE 5.12. In the population-genetic approach to the formula for the  $\theta$ -correction, we have looked at the probability that alleles in the population are IBD. When we follow the lineages of the alleles back in time, the “waiting times” for alleles to either coalesce or arise in the population as a result of immigration or mutation are exponentially distributed. Writing  $N$  for the population size, we assumed that the waiting time for an immigration or mutation event is exponentially distributed with parameter  $\frac{\mu}{2N}$ , whereas for any pair of lineages, the waiting time for coalescence is exponentially distributed with parameter  $\frac{1}{N}$ . For any two alleles, the probability of coalescence before mutation or immigration is then given by

$$\theta := \frac{\frac{1}{N}}{\frac{1}{N} + \frac{\mu}{N}}.$$

- (i) Consider three alleles. Show that the probability for a coalescence among these three alleles before any mutation/immigration event is equal to

$$\frac{\frac{3}{N}}{\frac{3\mu}{2N} + \frac{3}{N}}$$

and that this quantity is equal to  $\frac{2\theta}{1+\theta}$ .

- (ii) Show that the probability that all three alleles are IBD is equal to  $\frac{2\theta^2}{1+\theta}$ .  
 (iii) Show that the probability that none of the three alleles are IBD is equal to  $\frac{(1-\theta)^2}{1+\theta}$ .  
 (iv) Explain that the probability that all three alleles are  $a$  is equal to

$$\frac{2\theta^2 p_a}{1+\theta} + \frac{p_a^3(1-\theta)^2}{1+\theta} + \frac{3p_a^2(\theta-\theta^2)}{1+\theta},$$

where  $p_a$  is the population frequency of allele  $a$ .

SOLUTION. (i) There are  $\binom{3}{2} = 3$  possible pairs of alleles. The total rate for any pair to coalesce is  $\frac{3}{N}$ , and the total rate for any of the 3 lineages to undergo mutation/immigration is  $\frac{3\mu}{2N}$ . Therefore, the probability for a coalescence among these 3 alleles before any mutation/immigration is equal to  $\frac{\frac{3}{N}}{\frac{3\mu}{2N} + \frac{3}{N}}$ . Since

$$\theta = \frac{\frac{1}{N}}{\frac{1}{N} + \frac{\mu}{N}} \implies \mu = \frac{1-\theta}{\theta},$$

this quantity is  $\frac{2\theta}{1+\theta}$ .

- (ii) By (i), since the probability of coalescence before mutation/immigration for 2 lineages is  $\theta$ , the probability that all three alleles are IBD is  $\frac{2\theta^2}{1+\theta}$ .  
 (iii) The probability of mutation/immigration first for 3 lineages is

$$1 - \frac{2\theta}{1+\theta} = \frac{1-\theta}{1+\theta}.$$

The probability that mutation or immigration occurs first for 2 lineages is  $1-\theta$ ; therefore, the probability that none of the three alleles are IBD is  $\frac{(1-\theta)^2}{1+\theta}$ .

- (iv) We have the following cases:

- All three alleles are IBD: They share one common ancestor  $a$  with probability  $p_a$ . By (ii) we have  $\frac{2\theta^2}{1+\theta} p_a$ .
- None of the alleles are IBD: They are three independent draws from the population, and the probability that they are all of type  $a$  is  $p_a^3$ . By (iii) we have  $\frac{(1-\theta)^2}{1+\theta} p_a^3$ .
- Exactly two alleles are IBD: The probability of this state is

$$1 - \frac{2\theta^2}{1+\theta} - \frac{(1-\theta)^2}{1+\theta} = \frac{3(\theta-\theta^2)}{1+\theta}.$$

The probability that both lineages result in type  $a$  is  $p_a^2$ , giving  $\frac{3(\theta-\theta^2)}{1+\theta} p_a^2$ .

Combining these states yields the probability that all three alleles are  $a$  is  $\frac{2\theta^2 p_a}{1+\theta} + \frac{p_a^3(1-\theta)^2}{1+\theta} + \frac{3p_a^2(\theta-\theta^2)}{1+\theta}$ .  $\square$

# Statistical Modelling and DNA Mixture Evaluation

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## SECTION 6.1

## | Mixture Likelihood Ratios

Without drop-in ( $c = 0$ ) and  $n = 1$ , we have

$$P_\theta(a \in M \mid g) = 1 - \prod_{i=1}^n d_i^{n_{i,a}} = 1 - d^{n_a}, \quad (6.1)$$

where  $n_a$  is the number of alleles  $a$  in  $g$ . Moreover,

$$\begin{aligned} P(M \mid H, g, I) &= \int_0^1 \cdots \int_0^1 P_\theta(M \mid g_1, \dots, g_n) P(g_1, \dots, g_n \mid H, g, I) dx_1 \dots dx_n \\ &= \int_0^1 P_\theta(M \mid g) P(g \mid H, g, I) dx. \end{aligned} \quad (6.2)$$

## | Exercises

**EXERCISE 6.1.** A trace  $M$  is investigated on an autosomal locus. Alleles  $\{a, b\}$  are observed. Denote this by  $M = \{a, b\}$ . The contributors to the trace are a victim and a second person. The victim has genotype  $G_1 = (a, a)$ . We assume that the victim has dropout probability equal to zero. The dropout probability of the second contributor is some fixed  $d$  with  $1 > d \geq 0$ . We do not consider drop-in to be possible. Let  $G_2$  be the (random variable modelling the) genotype of the second contributor. Show that, under these assumptions,

$$\mathbb{P}(G_2 = (a, b) \mid M = \{a, b\}, G_1 = (a, a)) = \frac{2p_a}{(1-d)(2p_a + p_b) + 2d}.$$

**SOLUTION.** The possible genotypes for  $G_2$  are:

- $(a, b)$ :  $2p_a p_b (1-d)$ .
- $(b, b)$ :  $p_b^2 (1-d^2)$ .
- $(b, c)$  where  $c \notin \{a, b\}$ :  $\sum_{c \notin \{a, b\}} 2p_b p_c \cdot d(1-d)$ .

Therefore

$$\begin{aligned}
 \mathbb{P}(G_2 = (a, b) \mid M, G_1) &= \frac{2p_a p_b (1-d)}{2p_a p_b (1-d) + p_b^2 (1-d^2) + \sum_{c \notin \{a, b\}} 2p_b p_c d (1-d)} \\
 &= \frac{2p_a}{2p_a + p_b(1+d) + \sum_{c \notin \{a, b\}} 2p_c d} \\
 &= \frac{2p_a}{2p_a + p_b(1+d) + 2(1-p_a-p_b)d} = \frac{2p_a}{(1-d)(2p_a + p_b) + 2d}.
 \end{aligned}$$

□

EXERCISE 6.2. A DNA profile of a crime stain is generated, producing a single source profile that may be incomplete. We calculate likelihood ratios with the discrete model, ignoring drop-in. That is, if the contributor has  $n_a$  alleles of type  $a$  and the probability of dropout for the contributor is  $x$ , then the probability that allele  $a$  is detected in the trace profile is  $1 - x^{n_a}$ . We assume that the dropout probability  $x$  is uniformly distributed between 0 and 1. We consider information on two loci. On the first locus, the trace profile shows only allele  $a_1$ . On the second locus, alleles  $b_1$  and  $b_2$  are detected.

Calculate, based on these two loci, the probability distribution of the genotypes of the contributor to this trace. You may assume that on the first locus, the alleles occurring in the population are  $a_1$  and  $a_2$ , with known allele frequencies  $p_1$  and  $p_2$ , respectively; on the second locus, the alleles occurring in the population are  $b_1$ ,  $b_2$ , and  $b_3$ , with known allele frequencies  $q_1$ ,  $q_2$ , and  $q_3$ , respectively.

SOLUTION. Since we consider drop-in to be impossible, the possible genotypes of the contributor are either  $(a_1, a_1)$  or  $(a_1, a_2)$  on locus one, and with certainty  $(b_1, b_2)$  on locus two. Let

$$g_1 = ((a_1, a_1), (b_1, b_2)), \quad g_2 = ((a_1, a_2), (b_1, b_2)).$$

Then

$$\mathbb{P}(g_1 \mid E, x) = (1-x^2)(1-x)^2, \quad \mathbb{P}(g_2 \mid E, x) = x(1-x)^3.$$

We calculate

$$\begin{aligned}
 \mathbb{P}(E \mid g_1) &= \int_0^1 (1-x^2)(1-x)^2 dx = \frac{3}{10}, \\
 \mathbb{P}(E \mid g_2) &= \int_0^1 x(1-x)^3 dx = \frac{1}{20}.
 \end{aligned}$$

Hence,

$$\begin{aligned}
 \mathbb{P}(g_1 \mid E) &= \frac{\mathbb{P}(E \mid g_1)\mathbb{P}(g_1)}{\mathbb{P}(E \mid g_1)\mathbb{P}(g_1) + \mathbb{P}(E \mid g_2)\mathbb{P}(g_2)} \\
 &= \frac{\frac{3}{10}p_1^2(2q_1q_2)}{\frac{3}{10}p_1^2(2q_1q_2) + \frac{1}{20}(2p_1p_2)(2q_1q_2)} = \frac{3p_1}{2p_1 + 1}. \\
 \mathbb{P}(g_2 \mid E) &= 1 - \mathbb{P}(g_1 \mid E) = \frac{1-p_1}{2p_1 + 1}.
 \end{aligned}$$

□

EXERCISE 6.3. Consider the discrete model with integration of (6.1)-(6.2) without drop-in, i.e.,  $c = 0$ . We have a trace profile left by a single contributor ( $n = 1$ ), determined on two loci. On the first locus, only allele  $a$  is detected. On the second locus, no alleles are detected. There is a person of interest (PoI) with genotype  $(a, b)$  on the first locus, and genotype  $(c, d)$  on the

second. Denote the allele frequencies as  $p_a, p_b$  on the first locus and  $q_c, q_d$  on the second locus.

- (i) Calculate the likelihood ratio on the first locus only.
  - $H_1$ : The PoI is the contributor to the trace.
  - $H_2$ : An unknown person (unrelated to the PoI) is the contributor.
- (ii) Do the same, on the second locus only.
- (iii) Now calculate the likelihood ratio taking both loci into account.

SOLUTION. (i) Under  $H_1$ , given that  $M_1 = \{a\}$  and  $g_1 = (a, b)$ , the allele  $a$  was detected and allele  $b$  dropped out. Then  $P(M_1 | H_1, d) = (1 - d)d$ . Integrating over  $d$  gives:

$$P(M_1 | H_1) = \int_0^1 (d - d^2) dd = \frac{1}{6}.$$

Under  $H_2$ , since  $c = 0$ , the contributor must have allele  $a$ .

- When the genotype of the unknown person is  $g_U = (a, a)$  with probability  $p_a^2$ ,  $P(M_1 | g_U, d) = 1 - d^2$ .
- When  $g_U = (a, k)$ , where  $k \neq a$  with probability  $2p_a p_k$ ,  $P(M_1 | g_U, d) = d(1 - d)$ .

Combining these, we have

$$P(M_1 | H_2, d) = p_a^2(1 - d^2) + 2p_a(1 - p_a)(d - d^2).$$

Integrating over  $d$  gives:

$$\begin{aligned} P(M_1 | H_2) &= \int_0^1 p_a^2(1 - d^2) + 2p_a(1 - p_a)(d - d^2) dd \\ &= p_a^2 \int_0^1 (1 - d^2) dd + 2p_a(1 - p_a) \int_0^1 (d - d^2) dd \\ &= p_a^2 \left(1 - \frac{1}{3}\right) + 2p_a(1 - p_a) \left(\frac{1}{2} - \frac{1}{3}\right) \\ &= \frac{2}{3}p_a^2 + \frac{1}{3}p_a(1 - p_a) = \frac{1}{3}p_a(1 + p_a). \end{aligned}$$

Therefore, the likelihood ratio for locus 1 is  $\frac{1}{2p_a(1+p_a)}$ .

- (ii) We have  $M_2 = \emptyset$  and  $g_2 = (c, d)$ . Under  $H_1$ , both alleles dropped out and  $P(M_2 | H_1, d) = d^2$ . Under  $H_2$ , regardless of the genotype of U, all alleles dropped out again and  $P(M_2 | H_2, d) = d^2$ . Therefore, the likelihood ratio for locus 2 is 1.
- (iii)

$$P(M | H_1) = \int_0^1 P(M_1 | H_1, d)P(M_2 | H_1, d) dd = \int_0^1 (d^3 - d^4) dd = \frac{1}{20}.$$

$$\begin{aligned} P(M | H_2) &= \int_0^1 P(M_1 | H_2, d)P(M_2 | H_2, d) dd \\ &= \int_0^1 (p_a^2(d^2 - d^4) + 2p_a(1 - p_a)(d^3 - d^4)) dd = \frac{p_a(p_a + 3)}{30}. \end{aligned}$$

Therefore, the total likelihood ratio is  $\frac{3}{2p_a(p_a+3)}$ .

□

# $p$ -Values of Likelihood Ratios

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If we let  $H_2$  play the role of null hypothesis, then we can consider

$$\mathbb{P}(\text{LR}_{H_1, H_2}(E) \geq x \mid H_2)$$

as a  $p$ -value.

EXAMPLE 7.1 ( $p$ -Values May Change if the Distribution of Unobserved Possibilities Changes).

EXAMPLE 7.2 ( $p$ -Values are Ambiguous).

EXAMPLE 7.3 ( $p$ -Values May Appear to Support the Wrong Hypothesis).

## | Exercises

EXERCISE 7.1. Consider a locus with two alleles  $a_1, a_2$  that exist in the population and assume these are equally frequent. We will consider two-person mixtures without dropout or drop-in, where one contributor is supposed to be known (i.e., their genotype known) and the other is to be identified via LR calculations. Hence the hypotheses will be of the form “contributors are the known contributor (KC) and the person of interest (PoI)” versus “known contributor and one unknown individual (unrelated to PoI or KC)”. We work with the “combinatorial” likelihoods (i.e., consider the detected alleles as the evidence, and do not model dropout and drop-in).

- (i) Argue that when the known contributor has genotype  $(a_1, a_2)$ , no evidence can be obtained.
- (ii) When the known contributor is homozygous, give the possible resulting mixtures and LRs for each genotype of a person of interest.
- (iii) Give the probability distribution  $\mathbb{P}(\text{LR} = x \mid H_1)$  where  $H_1$  (stating that the known contributor and the PoI contributed) is compared to  $H_2$  (stating that the known contributor and one unknown unrelated person contributed) prior (i.e., unconditional on) to obtaining the mixed profile and the known contributor’s genotype.
- (iv) Now suppose that there are 10 such loci. Compute the probability that a likelihood ratio of at least  $10^4$  is obtained in the above setting when  $H_1$  is true.

SOLUTION. The hypotheses are:

- $H_1$ : The contributors are KC and PoI.
  - $H_2$ : The contributors are KC and an unrelated unknown person (U).
- (i) If KC has genotype  $(a_1, a_2)$ , then the mixture  $M$  must contain both alleles  $a_1$  and  $a_2$ , regardless of the other contributor, i.e.,  $M = \{a_1, a_2\}$ . The likelihood ratio is then

$$\frac{P(M | H_1)}{P(M | H_2)} = 1,$$

which implies the evidence provides equal support for both hypotheses, and hence, no evidence can be obtained.

- (ii) Given that  $P(a_1) = P(a_2) = \frac{1}{2}$ , without loss of generality, we assume the homozygous KC has  $g_{KC} = (a_1, a_1)$ . Then for PoI genotypes:

- $g_{PoI} = (a_1, a_1)$ :  $M = \{a_1\}$ ,

$$\frac{P(M | H_1)}{P(M | g_{KC}, g_U = (a_1, a_1))} = \frac{1}{0.5^2} = 4.$$

- $g_{PoI} = (a_1, a_2)$ :  $M = \{a_1, a_2\}$ ,

$$\frac{P(M | H_1)}{P(M | g_{KC}, g_U = \{(a_1, a_2), (a_2, a_2)\})} = \frac{1}{0.5^2 + 2 \times 0.5 \times 0.5} = \frac{4}{3}.$$

- $g_{PoI} = (a_2, a_2)$ :  $M = \{a_1, a_2\}$ ,

$$\frac{P(M | H_1)}{P(M | g_{KC}, g_U = \{(a_1, a_2), (a_2, a_2)\})} = \frac{1}{0.5^2 + 2 \times 0.5 \times 0.5} = \frac{4}{3}.$$

- (iii) When KC is heterozygous with  $g_{KC} = (a_1, a_2)$ , as in part (i),

$$P(LR = 1 | H_1) = \frac{1}{2}.$$

When KC is homozygous, we again assume  $g_{KC} = (a_1, a_1)$ . By part (ii):

- $g_{PoI} = (a_1, a_1)$ :

$$P(LR = 4 | H_1) = \frac{1}{2} \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{8}.$$

- $g_{PoI} = \{(a_1, a_2), (a_2, a_2)\}$ :

$$P\left(LR = \frac{4}{3} | H_1\right) = \frac{1}{2} \cdot \left(\left(\frac{1}{2}\right)^2 + 2 \cdot \frac{1}{2} \cdot \frac{1}{2}\right) = \frac{3}{8}.$$

- (iv) Let  $N_1$ ,  $N_{\frac{4}{3}}$ , and  $N_4$  be the number of loci where the likelihood ratio is 1,  $\frac{4}{3}$ , and 4, respectively.

Then  $N_1 + N_{\frac{4}{3}} + N_4 = 10$ , and the total likelihood ratio is  $4^{N_4} \left(\frac{4}{3}\right)^{N_{\frac{4}{3}}} \cdot 1^{N_1}$ . We need to find the probability that

$$4^{N_4} \left(\frac{4}{3}\right)^{N_{\frac{4}{3}}} \geq 10000.$$

Taking the logarithm on both sides,

$$N_4 \log 4 + N_{\frac{4}{3}} (\log 4 - \log 3) \geq 4.$$

Using  $\log 3 \approx 0.477$  and  $\log 4 \approx 0.602$ ,

$$0.602N_4 + 0.125N_{\frac{4}{3}} \geq 4.$$

Observe that when  $N_4 = 5$ ,  $N_{\frac{4}{3}} \geq 7.92$ , which implies the minimum  $N_{\frac{4}{3}}$  is 8, and the total number of loci exceeds 10. When  $N_4 = 6$ ,  $N_{\frac{4}{3}} \geq 3.104$ , and  $N_{\frac{4}{3}} = 4$  satisfies the condition. We also verify that when  $N_4 \geq 7$ , the condition is satisfied. Then we consider the following two cases:

–  $N_4 \geq 7$ :

$$P(N_4 = \{7, 8, 9, 10\}) = \sum_{k=7}^{10} \binom{10}{k} \left(\frac{1}{8}\right)^k \left(1 - \frac{1}{8}\right)^{10-k}.$$

Plugging in the numbers, we have

$$P(N_4 = 7) = 120 \cdot \frac{1}{8^7} \cdot \frac{343}{8^3} = \frac{41160}{8^{10}},$$

$$P(N_4 = 8) = 45 \cdot \frac{1}{8^8} \cdot \frac{49}{8^2} = \frac{2205}{8^{10}},$$

$$P(N_4 = 9) = 10 \cdot \frac{1}{8^9} \cdot \frac{7}{8^1} = \frac{70}{8^{10}},$$

$$P(N_4 = 10) = 1 \cdot \frac{1}{8^{10}} \cdot 1 = \frac{1}{8^{10}}.$$

These add up to  $\frac{43436}{8^{10}}$ .

–  $N_4 = 6$  and  $N_{\frac{4}{3}} = 4$ :

$$P(N_4 = 6) = \frac{10!}{6!4!0!} \left(\frac{1}{8}\right)^6 \left(\frac{3}{8}\right)^4 \left(\frac{1}{2}\right)^0 = 210 \cdot \frac{1}{8^6} \cdot \frac{81}{8^4} = \frac{17010}{8^{10}}.$$

Therefore, the total probability is

$$\frac{43436 + 17010}{8^{10}} = \frac{60446}{1073741824} \approx 5.629 \times 10^{-5}.$$

□

# The Interpretation of DNA Database Matches

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## SECTION 8.1

## | The Database Controversy

A database contains “reference” profiles of convicted offenders and suspects. Given a profile  $g_C$  of an unknown offender  $C$  (here from a single source rather than a mixture), compare two scenarios. Under “probable cause”, a suspect  $S$  is identified first and a “match” is then found, i.e.,  $g_S = g_C$ . Under a “cold hit”, no suspect is identified through the investigation; instead,  $S$  is found because this person is the only individual in the database with a matching profile. Does the evidential value differ? When the debate started, likelihood ratios were rarely used; random match probabilities were used instead.

- The random match probability still applies to a “cold hit”. Moreover, many people can no longer be  $C$  (all those in  $D$  except  $S$ )  $\rightarrow$  stronger evidence.
- Random match probability does not apply since any match in  $D$  would lead to a suspect  $\rightarrow$  weaker evidence for “cold hit”.

Initially, in 1992, a report of the *National Research Council* (NRC-I) advised that the match not be treated as evidence in court and that it be used only to identify the matching individual as a suspect. In 1996, NRC-II recommended multiplying the *random match probability* by the number  $n$  of persons in the database.

$E = \{g_S = g\} \cup E'$ ,  $G_C = g$ , where  $E'$  denotes no other matches in  $D$ . The likelihood ratio

$$\frac{\mathbb{P}(G_S = g, E' \mid G_C = g, C = S)}{\mathbb{P}(G_S = g, E' \mid G_C = g, C \neq S)} = \frac{\mathbb{P}(G_S = g \mid G_C = g, C = S) \mathbb{P}(E' \mid G_C = g, G_S = g, C = S)}{\mathbb{P}(G_S = g \mid G_C = g, C \neq S) \mathbb{P}(E' \mid G_C = g, G_S = g, C \neq S)}$$

is greater than  $\frac{1}{p}$  since  $E' \Rightarrow C \notin D \setminus \{S\}$ .

## SECTION 8.2

## | Mathematical Modelling and the Basic Formula

The posterior odds of  $C = i$  versus  $C \neq i$  are given by

$$\frac{\mathbb{P}(C = i | E_i)}{\mathbb{P}(C \neq i | E_i)} = \frac{1}{p_i} \frac{\mathbb{P}(C = i)}{\mathbb{P}(C \notin \mathcal{D})}. \quad (8.1)$$

## | Exercises

EXERCISE 8.1. A trace profile from an unknown donor  $C$  is compared with a DNA database  $\mathcal{D}$  that contains the DNA profiles of  $n = 10^6$  persons. Prior to the search, there is no clue as to the identity of the trace donor  $C$ . It is assessed that  $\mathbb{P}(C \in \mathcal{D}) = \frac{1}{11}$ . A single match is obtained with person  $S$ ; this match has random match probability  $10^{-9}$ .

- (i) Give the posterior odds for  $C = S$  versus  $C \neq S$  based on this information.
- (ii) Now suppose that there is one locus for which we know the genotype of  $C$ , but not of  $S$ . The genotype of  $C$  at this locus occurs in 10% of the population. What is the probability that, if we also determined the genotype of  $S$  at this locus, we would find that  $S$  has the same genotype as  $C$ ?

EXERCISE 8.2. In a database (in which all persons are typed on the same loci) we search for  $C$ , an unknown trace donor. The database contains one matching person, that we call  $S$ . We write this result as  $E_S$ . The posterior odds on  $S = C$  are

$$\frac{\mathbb{P}(S = C | E_S)}{\mathbb{P}(S \neq C | E_S)} = \frac{1}{p} \frac{\mathbb{P}(S = C)}{\mathbb{P}(C \notin \mathcal{D})}.$$

- (i) Explain the meaning of the terms on the right-hand side.
- (ii) It is known that when  $p = 10^{-7}$ , 90% of all matches are with the trace donors. How much is that for  $p = 10^{-8}$ ?
- (iii) One year after the match with  $S$  was found, the database is searched again. It has grown since the last search. Call the new database  $\mathcal{D}'$ , an extension of  $\mathcal{D}$ . In  $\mathcal{D}'$ ,  $S$  is still the only match. Which of the probabilities on the right-hand side have changed, and how? What is the effect on the posterior odds?
- (iv) Suppose all persons in the database are equally likely to be trace donors and that the prior odds  $\frac{\mathbb{P}(C \in \mathcal{D})}{\mathbb{P}(C \notin \mathcal{D})} = 0.9$ . How many persons are in the database?

EXERCISE 8.3. We carry out a familial search to identify the father  $F$  of an unknown offender in a database  $\mathcal{D}$  consisting of  $N = 500,000$  persons. We estimate that  $\mathbb{P}(F \in \mathcal{D}) = \frac{1}{11}$ . Let  $E$  denote the event that for the individuals in  $\mathcal{D}$  the likelihood ratios in favour of being  $F$ , versus being unrelated to  $F$ , are  $r_1 = 1,000,000$ ,  $r_2 = 500,000$  and  $\sum_{i=3}^N r_i = 500,000$ .

- (i) Calculate  $\mathbb{P}(F \in \mathcal{D} | E)$  and  $\mathbb{P}(F = 1 | E)$ .
- (ii) If it now became known that  $F \neq 2$ , what would be the impact on the probabilities that  $F \in \mathcal{D}$  and that  $F = 1$ ?

EXERCISE 8.4. In a database  $\mathcal{D}$  with the DNA profiles of 300,000 individuals, we look for the donor  $C$  of a crime stain from which a DNA profile  $g_S$  has been obtained. It is believed that the probability that  $\mathcal{D}$  contains  $C$  is  $\mathbb{P}(C \in \mathcal{D}) = 0.6$ , but there is a priori no clue as to who in the database might be  $C$ . A single match with  $S \in \mathcal{D}$  is found upon comparison. If  $S$  were not  $C$ , the probability that their profiles would match on all compared loci is  $10^{-9}$ . We denote these findings by  $E_S$ . We have

$$\frac{\mathbb{P}(C = S \mid E_S)}{\mathbb{P}(C \neq S \mid E_S)} = \frac{1}{p_S} \frac{\mathbb{P}(C = S)}{\mathbb{P}(C \notin \mathcal{D})}.$$

- (i) Show that the posterior odds

$$\frac{\mathbb{P}(C = S \mid E_S)}{\mathbb{P}(C \neq S \mid E_S)} = 5000.$$

- (ii) Determine the prior odds, likelihood ratio and posterior odds for hypotheses  $C \in \mathcal{D}$  versus  $C \notin \mathcal{D}$  and evidence  $E_S$ .  
 (iii) Determine the prior odds, likelihood ratio and posterior odds for hypotheses  $C = S$  versus  $C \neq S$  and evidence  $E_S$ .  
 (iv) Now suppose the situation were different, so that  $S$  had a large a priori probability of being  $C$ , but  $\mathbb{P}(C \in \mathcal{D} \mid C \neq S)$  remained the same as before. Would the likelihood ratio in (ii) change and, if so, in which direction?

SOLUTION. (i) Use  $\mathbb{P}(C = S) = \frac{1}{300000} \mathbb{P}(C \in \mathcal{D}) = 2 \times 10^{-6}$ .

- (ii) The posterior odds are still 5000 since  $C \in \mathcal{D}$  and  $C = S$  are conditionally equivalent given  $E_S$ . The prior odds are  $\frac{0.6}{0.4} = \frac{3}{2}$ , so the LR is  $\frac{10000}{3}$ .  
 (iii) The posterior odds are again 5000 and the prior odds are  $\frac{2 \times 10^{-6}}{1 - 2 \times 10^{-6}} = \frac{1}{499999}$ ; this gives  $\text{LR} \approx 2.5 \times 10^9$ .  
 (iv) The likelihood ratio in (ii) is equal to

$$\frac{\mathbb{P}(C = S \mid C \in \mathcal{D})(1 - p_S)^{N-1}}{p_S(1 - p_S)^{N-1}} = \frac{1}{300000} \times 10^9.$$

If  $C = S$  is a priori more likely, then  $\mathbb{P}(C = S \mid C \in \mathcal{D})$  will be larger, and so will the likelihood ratio. □

EXERCISE 8.5. In a DNA database  $\mathcal{D}$  we search for  $C$ , the unknown donor of a trace. We assume that the population of potential trace donors consists of  $N = 10^7$  persons. These are added to the database (and never removed) in such an order that when the database contains the profiles of  $n$  (out of these  $N$ ) persons,

$$\mathbb{P}(C \in \mathcal{D}) = \sqrt{\frac{n}{N}}.$$

We also assume that for all persons in the database, the profiles are determined on the same set of loci so that all database members have the same random match probability  $p$ .

When  $n = 10^5$ , we search for  $C$ , the donor of a trace with random match probability  $p = 10^{-8}$  for the loci that the database contains. We find no match. We let the database grow over time and the search is re-run when  $n = 4 \cdot 10^5$ . Now there is a unique match in the database.

- (i) What are the odds on this match being with the trace donor?

- (ii) Later, when the database has grown to  $n = 9 \cdot 10^5$ , the search is run once again but no new matches are obtained. How, if at all, does this change the odds on the trace donor being with the previously identified matching individual  $S$ ?

SOLUTION. (i) Let  $\mathcal{D}$  denote the database with 400,000 persons. Then there is a unique match in  $\mathcal{D}$  which is with a person in  $\mathcal{D}'$ , where  $\mathcal{D}'$  is the sub-database with the last 300,000 persons. We have

$$\mathbb{P}(C \in \mathcal{D}) = 0.2, \quad \mathbb{P}(C \in \mathcal{D}') = 0.1, \quad \mathbb{P}(C \in \mathcal{D} \setminus \mathcal{D}') = 0.1.$$

Let  $E_1$  denote that there is no match among the first 100,000 persons, and  $E_2$  that there is a unique match among the next 300,000. Then  $E_1$  rules out that  $C \in \mathcal{D} \setminus \mathcal{D}'$  and we have  $\mathbb{P}(C \in \mathcal{D} \setminus \mathcal{D}' | E_1) = \frac{0.1}{0.9} = \frac{1}{9}$ . The odds on the match being with the trace donor are now

$$\frac{\mathbb{P}(C = S | E_2)}{\mathbb{P}(C \neq S | E_2)} = \frac{1}{n_p} \frac{\mathbb{P}(C \in \mathcal{D} \setminus \mathcal{D}' | E_1)}{\mathbb{P}(C \notin \mathcal{D} \setminus \mathcal{D}' | E_1)} = \frac{1}{300000 \times 10^{-8}} \cdot \frac{\frac{1}{9}}{\frac{8}{9}} = \frac{125}{3}.$$

□

EXERCISE 8.6. Suppose that we have a database  $\mathcal{D}$  with the DNA profiles of individuals, and we compare them with the DNA profile  $G_C$  of an unknown offender  $C$ . The search excludes all but the persons  $S_1, S_2, \dots, S_k \in \mathcal{D}$  for some  $k \geq 1$ . We remark that the profiles of the matching individuals  $S_i$  need not be on the same loci, and we let  $p_i$  be the probability that  $S_i$  would match on the loci compared between  $S_i$  and  $C$ , if  $C \neq S_i$ . We summarize the obtained evidence by  $E$ .

- (i) Show that

$$\mathbb{P}(C = S_1 | E) = \frac{\mathbb{P}(C = S_1)/p_1}{\mathbb{P}(C \notin \mathcal{D}) + \sum_{i=1}^k \mathbb{P}(C = S_i)/p_i}.$$

- (ii) Show that the likelihood ratio for  $C \in \mathcal{D}$  versus  $C \notin \mathcal{D}$  of  $E$  is

$$\frac{\mathbb{P}(E | C \in \mathcal{D})}{\mathbb{P}(E | C \notin \mathcal{D})} = \sum_{i=1}^k \frac{\mathbb{P}(C = S_i | C \in \mathcal{D})}{p_i}.$$

- (iii) Now let all  $p_i := p$  be equal and let the distribution of  $C$  on  $\mathcal{D}$  be uniform. Express  $\mathbb{P}(C = S_i | E)$  in terms of  $N, p, k$  and prior probabilities.

SOLUTION. (i) Write  $\mathcal{D} = \{S_1, \dots, S_k\} \cup \mathcal{D}'$ . Let  $q$  be the probability of not obtaining any matches in  $\mathcal{D}'$  given that  $C \notin \mathcal{D}'$ . If  $C \notin \mathcal{D}$ , all matches with the  $S_i$  are coincidental:

$$\mathbb{P}(E | C \notin \mathcal{D}) = q \prod_{j=1}^k p_j.$$

Then

$$\mathbb{P}(E | C = S_1) = q \prod_{j=2}^k p_j, \quad \mathbb{P}(E | C = S_i) = q \prod_{j \neq i} p_j.$$

Therefore,

$$\begin{aligned} \mathbb{P}(C = S_1 | E) &= \frac{\mathbb{P}(E | C = S_1) \mathbb{P}(C = S_1)}{\mathbb{P}(E | C \notin \mathcal{D}) \mathbb{P}(C \notin \mathcal{D}) + \sum_{i=1}^k \mathbb{P}(E | C = S_i) \mathbb{P}(C = S_i)} \\ &= \frac{q \prod_{j=2}^k p_j \mathbb{P}(C = S_1)}{q \prod_{j=1}^k p_j \mathbb{P}(C \notin \mathcal{D}) + \sum_{i=1}^k q \prod_{j \neq i} p_j \mathbb{P}(C = S_i)} \end{aligned}$$

$$\begin{aligned}
&= \frac{\mathbb{P}(C = S_1)}{p_1 \mathbb{P}(C \notin \mathcal{D}) + \sum_{i=1}^k \mathbb{P}(C = S_i) \cdot \frac{\prod_{j \neq i} p_j}{\prod_{j=2}^k p_j}} \\
&= \frac{\mathbb{P}(C = S_1)}{p_1 \mathbb{P}(C \notin \mathcal{D}) + \mathbb{P}(C = S_1) + \sum_{i=2}^k \mathbb{P}(C = S_i) \frac{p_1}{p_i}} = \frac{\mathbb{P}(C = S_1)/p_1}{\mathbb{P}(C \notin \mathcal{D}) + \sum_{i=1}^k \mathbb{P}(C = S_i)/p_i}.
\end{aligned}$$

(ii) By (i),

$$\begin{aligned}
\mathbb{P}(E | C \in \mathcal{D}) &= \sum_{i=1}^k \mathbb{P}(E | C = S_i) \mathbb{P}(C = S_i | C \in \mathcal{D}) + \underbrace{\mathbb{P}(E | C \in \mathcal{D}') \mathbb{P}(C \in \mathcal{D}' | C \in \mathcal{D})}_{=0} \\
&= \sum_{i=1}^k \mathbb{P}(C = S_i | C \in \mathcal{D}) q \prod_{j \neq i} p_j.
\end{aligned}$$

This gives as likelihood ratio

$$\frac{\mathbb{P}(E | C \in \mathcal{D})}{\mathbb{P}(E | C \notin \mathcal{D})} = \frac{\sum_{i=1}^k \mathbb{P}(C = S_i | C \in \mathcal{D}) q \prod_{j \neq i} p_j}{q \prod_{j=1}^k p_j} = \sum_{i=1}^k \frac{\mathbb{P}(C = S_i | C \in \mathcal{D})}{p_i}.$$

(iii) The likelihood ratio reduces to

$$\sum_{i=1}^k \frac{\mathbb{P}(C = S_i | C \in \mathcal{D})}{p_i} = \sum_{i=1}^k \frac{1/N}{p} = \frac{k}{Np}.$$

The posterior odds on  $C \in \mathcal{D}$  become

$$\frac{\mathbb{P}(C \in \mathcal{D} | E)}{\mathbb{P}(C \notin \mathcal{D} | E)} = \frac{k}{Np} \cdot \frac{\mathbb{P}(C \in \mathcal{D})}{\mathbb{P}(C \notin \mathcal{D})},$$

hence the posterior probability is

$$\mathbb{P}(C \in \mathcal{D} | E) = \frac{\frac{k\mathbb{P}(C \in \mathcal{D})}{Np\mathbb{P}(C \notin \mathcal{D})}}{1 + \frac{k\mathbb{P}(C \in \mathcal{D})}{Np\mathbb{P}(C \notin \mathcal{D})}} = \frac{k\mathbb{P}(C \in \mathcal{D})}{k\mathbb{P}(C \in \mathcal{D}) + Np\mathbb{P}(C \notin \mathcal{D})}.$$

This probability is concentrated on the  $S_i$  since these are the only non-excluded persons in  $\mathcal{D}$ . All  $S_i$  must have the same posterior probability to be  $C$  since the prior for  $C$  on  $\mathcal{D}$  is uniform and all  $p_i$  are the same. Hence, for any  $i$ ,

$$\mathbb{P}(C = S_i | E) = \frac{\mathbb{P}(C \in \mathcal{D})}{k\mathbb{P}(C \in \mathcal{D}) + Np\mathbb{P}(C \notin \mathcal{D})}.$$

□

**EXERCISE 8.7.** A DNA database  $\mathcal{D}$  consists of two disjoint parts:  $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$ . The set of loci typed for the individuals in  $\mathcal{D}_1$  differs from the set of loci typed for the individuals in  $\mathcal{D}_2$ . A search is carried out with a case profile. In  $\mathcal{D}_1$ , the random match probability of this case profile is  $p_1$ ; in  $\mathcal{D}_2$ , it is  $p_2$ .

- (i) Suppose a single match is found with an individual in  $\mathcal{D}_1$ . Call this individual  $S_1$ . Let  $C$  be the person who left the trace. Express the posterior odds for  $H_1 : C = S_1$  versus  $H_2 : C \neq S_1$  in terms of the prior distribution of  $C$ ,  $p_1$  and  $p_2$ .<sup>1</sup>
- (ii) Suppose that, instead of the search result under (i), we have two matching individuals,  $S_1$  in  $\mathcal{D}_1$  and  $S_2$  in  $\mathcal{D}_2$ . Call this result  $E$ . Show that, using the same assumptions as

<sup>1</sup> We assume that there is no relatedness to take into account.

under (i),

$$\frac{\mathbb{P}(C = S_1 | E)}{\mathbb{P}(C \neq S_1 | E)} = \frac{p_2 \mathbb{P}(C = S_1)}{p_1 \mathbb{P}(C = S_2) + p_1 p_2 \mathbb{P}(C \notin \mathcal{D})}.$$

(iii) Under (ii), show that

$$\frac{\mathbb{P}(C \in \mathcal{D} | E)}{\mathbb{P}(C \notin \mathcal{D} | E)} = \frac{1}{p_1} \cdot \frac{\mathbb{P}(C = S_1)}{\mathbb{P}(C \notin \mathcal{D})} + \frac{1}{p_2} \cdot \frac{\mathbb{P}(C = S_2)}{\mathbb{P}(C \notin \mathcal{D})}.$$

SOLUTION. (i) Let part  $\mathcal{D}_i$  have  $N_i$  profiles, and let  $E$  denote the event that a single match with  $S_1$  in  $\mathcal{D}_1$  is obtained. Then

$$\begin{aligned}\mathbb{P}(E | C = S_1) &= (1 - p_1)^{N_1-1} (1 - p_2)^{N_2}, \\ \mathbb{P}(E | C \neq S_1) &= \mathbb{P}(C \notin \mathcal{D} | C \neq S_1) p_1 (1 - p_1)^{N_1-1} (1 - p_2)^{N_2}.\end{aligned}$$

Hence, the likelihood ratio is

$$\frac{1}{p_1 \mathbb{P}(C \notin \mathcal{D} | C \neq S_1)} = \frac{1}{p_1 \mathbb{P}(C \notin \mathcal{D} | C \neq S_1)}.$$

The posterior odds are

$$\frac{\mathbb{P}(C = S_1 | E)}{\mathbb{P}(C \neq S_1 | E)} = \frac{\mathbb{P}(C \neq S_1)}{p_1 \mathbb{P}(C \notin \mathcal{D})} \cdot \frac{\mathbb{P}(C = S_1)}{\mathbb{P}(C \neq S_1)} = \frac{\mathbb{P}(S = C)}{p_1 \mathbb{P}(C \notin \mathcal{D})}.$$

(ii) Now

$$\mathbb{P}(E | C = S_1) = (1 - p_1)^{N_1-1} p_2 (1 - p_2)^{N_2-1}.$$

If  $C \neq S_1$ , the result is only possible if  $C = S_2$  or  $C \notin \mathcal{D}$ . Thus,

$$\begin{aligned}\mathbb{P}(E | C \neq S_1) &= p_1 (1 - p_1)^{N_1-1} p_2 (1 - p_2)^{N_2-1} \mathbb{P}(C = S_2 | C \neq S_1) \\ &\quad + p_1 (1 - p_1)^{N_1-1} p_2 (1 - p_2)^{N_2-1} \mathbb{P}(C \notin \mathcal{D} | C \neq S_1).\end{aligned}$$

Since

$$\mathbb{P}(C = S_2 | C \neq S_1) = \frac{\mathbb{P}(C = S_2)}{\mathbb{P}(C \neq S_1)}$$

and similarly for the other term,

$$\text{LR} = \frac{p_2 \mathbb{P}(C \neq S_1)}{p_1 \mathbb{P}(C = S_2) + p_1 p_2 \mathbb{P}(C \notin \mathcal{D})}.$$

Now multiply this with the prior odds.

□

# Appendix

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EXERCISE A.1. Consider two hypotheses  $H_1$  and  $H_2$  and write  $\text{LR}(E)$  to denote the likelihood ratio  $\frac{\mathbb{P}(E|H_1)}{\mathbb{P}(E|H_2)}$ . Write

$$\alpha(t) := \mathbb{P}(\text{LR}(E) \leq t \mid H_1), \quad \beta(t) := \mathbb{P}(\text{LR}(E) > t \mid H_2).$$

(i) Show that

$$\alpha(t) + \beta(t) = 1 + \sum_{E: \text{LR}(E) \leq t} (\mathbb{P}(E \mid H_1) - \mathbb{P}(E \mid H_2)). \quad (\text{A.1})$$

(ii) Show that (A.1) is minimal when  $t = 1$ .

SOLUTION. (i) By definition,

$$\alpha(t) = \sum_{E: \text{LR}(E) \leq t} \mathbb{P}(E \mid H_1), \quad \beta(t) = \sum_{E: \text{LR}(E) > t} \mathbb{P}(E \mid H_2) = 1 - \sum_{E: \text{LR}(E) \leq t} \mathbb{P}(E \mid H_2).$$

Therefore

$$\alpha(t) + \beta(t) = \sum_{E: \text{LR}(E) \leq t} \mathbb{P}(E \mid H_1) + 1 - \sum_{E: \text{LR}(E) \leq t} \mathbb{P}(E \mid H_2) = 1 + \sum_{E: \text{LR}(E) \leq t} (\mathbb{P}(E \mid H_1) - \mathbb{P}(E \mid H_2)).$$

(ii) Define

$$f(t) := \alpha(t) + \beta(t) = 1 + \sum_{E: \text{LR}(E) \leq t} (\mathbb{P}(E \mid H_1) - \mathbb{P}(E \mid H_2)).$$

Notice that

$$\mathbb{P}(E \mid H_1) - \mathbb{P}(E \mid H_2) = \mathbb{P}(E \mid H_2)(\text{LR}(E) - 1).$$

When  $t < 1$ ,

$$f(1) - f(t) = \sum_{E: t < \text{LR}(E) \leq 1} (\mathbb{P}(E \mid H_1) - \mathbb{P}(E \mid H_2)) \leq 0.$$

When  $t > 1$ ,

$$f(t) - f(1) = \sum_{E: 1 < \text{LR}(E) \leq t} (\mathbb{P}(E \mid H_1) - \mathbb{P}(E \mid H_2)) \geq 0.$$

Therefore,  $f(t) \geq f(1)$  for all  $t$ , so  $\alpha(t) + \beta(t)$  is minimised at  $t = 1$ . □

EXERCISE A.2. For each of the following statements, decide whether it is true or not and motivate your choice.

- (i) Suppose two inbred individuals have a child. Then the child is also inbred.
- (ii) It is possible that  $\text{LR}_{H_1, H_2}(E) < 1$  but  $\mathbb{P}(H_1 \mid E) > \mathbb{P}(H_1)$ .
- (iii) It is known that even a perfect fingerprint can in itself never lead to a likelihood ratio larger than 1,000. This means that the strongest fingerprint evidence is stronger than DNA evidence with the same likelihood ratio.

- SOLUTION. (i) No. Someone is inbred means that their parents are related.  
(ii) True.

$$\mathbb{P}(H_1) = \mathbb{P}(H_2) = 0.1, \quad \mathbb{P}(H_3) = 0.8,$$

$$\text{LR}_{H_1, H_2}(E) = \frac{1}{2}, \quad \text{LR}_{H_1, H_3}(E) = 8.$$

Then

$$\mathbb{P}(H_1 | E) = \frac{\mathbb{P}(E | H_1)\mathbb{P}(H_1)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E | H_1)\mathbb{P}(H_1)}{\sum_i \mathbb{P}(E | H_i)\mathbb{P}(H_i)} = \frac{1}{4}.$$

- (iii) No. A likelihood ratio has the same meaning regardless of the evidence type.

□

EXERCISE A.3. When forensic DNA profiles of traces are interpreted, errors are sometimes made in the following sense. Let  $C$  be an unknown criminal who is the donor of a stain. We let  $G_C^{\text{lab}}$  be the DNA profile attributed to  $C$  by the forensic laboratory examining the stain. If  $G_C^{\text{lab}} = g'$  and  $G_C = g$  for  $g' \neq g$ , then an error has been made. We assume that  $\mathbb{P}(G_C^{\text{lab}} = g | G_C = g)$  is independent of the profile  $g$  and that  $\mathbb{P}(G_C^{\text{lab}} = g) = \mathbb{P}(G_C = g)$ . Furthermore, we assume that errors are made independently of whether  $S = C$ . Now let  $E = \{G_C^{\text{lab}} = g, G_S = g\}$  be the event that the profiles registered for the suspect  $S$  and the criminal  $C$  are both equal to  $g$ . Let  $Z$  be the indicator variable for an error:  $Z = 0$  if  $G_C^{\text{lab}} = G_C$  and  $Z = 1$  otherwise.

- (i) Show that the likelihood ratio in favour of  $Z = 0$  versus  $Z = 1$  is equal to

$$1 + \frac{\delta}{p_g(1 - \delta)},$$

where  $\delta = \mathbb{P}(S = C)$  and  $p_g$  is the population frequency of  $g$  (you may assume that DNA profiles of different persons are independent).

- (ii) Explain the limiting values of the likelihood ratio in (i) when  $\delta \downarrow 0$  and when  $\delta \uparrow 1$ .  
(iii) Derive the posterior probability  $\mathbb{P}(Z = 0 | E)$  and show that this probability tends to 1 if  $p_g \rightarrow 0$ . Can you explain this?

EXERCISE A.4. Show that

$$\frac{\mathbb{P}(H_1 | \text{LR}(E) \geq t)}{\mathbb{P}(H_2 | \text{LR}(E) \geq t)} = \mathbb{E}(\text{LR} | \text{LR} \geq t, H_2) \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_2)}.$$

EXERCISE A.5. Suppose that the population frequency of a certain trait is unknown to us. Suppose that our knowledge at this point is that the population frequency has distribution  $W$  with density  $f_W(t)$ , for  $t \in [0, 1]$ . Assume that  $\mathbb{E}[W] = p$  and that the variance of  $W$  is  $\sigma^2$ .

- (i) We select a random person from the population. Show that the probability that this person has the trait is  $p$ .  
(ii) Condition now on the first chosen person not having the trait. Compute the density of  $W$  conditional on this event and the corresponding (conditional) expectation.  
(iii) Now let  $W$  have density 1 on  $[0, 1]$ . Condition on the first  $n$  persons all having the trait. Show that the expectation of  $W$  conditioned on this tends to 1 as  $n \rightarrow \infty$ . Also explain this without any computation.

EXERCISE A.6. A crime is committed and a DNA profile, belonging with certainty to the unknown offender, is obtained. The police have a person of interest identified but cannot obtain DNA information because the evidence is at this point insufficient to arrest this person as suspect, and they need to first gather additional information. For this purpose, the offender's DNA is compared to that of the suspect's brother  $B$ , who happens to be in the DNA database. A forensic laboratory has computed a likelihood ratio for the following hypotheses:

- $H_p$ : The unknown offender is the brother of the database person  $B$ .
- $H_d$ : The unknown offender is unrelated to the database person  $B$ .

The likelihood ratio turns out to be 1. Furthermore, the forensic laboratory has calculated (and you can assume that this has been done correctly) that only one in ten thousand unrelated individuals has a likelihood ratio of at least 1 in favour of being a brother of  $B$ .

- (i) The forensic report now claims that there is a probability of one in ten thousand that the unknown offender is not the brother of  $B$ . Is this correct?
- (ii) If the report had stated that there is a probability of one in ten thousand that the unknown offender is unrelated to  $B$ , would this have been a correct statement?

EXERCISE A.7. A person and a trace are homozygous for allele 15 on locus D18S51. Suppose that in the reference database, which contains the genotypes of 250 persons, allele 15 occurs in 2 persons who are homozygous and 46 persons who are heterozygous.

- (i) Give the formula for the match probability, with  $\theta$ -correction, that applies here. (Recall that one uses the Dirichlet distribution here, starting with parameters  $\left(\frac{(1-\theta)p_1}{\theta}, \dots, \frac{(1-\theta)p_n}{\theta}\right)$  for appropriate  $p_1, \dots, p_n$ .)
- (ii) What is the frequency of allele 15 in the reference database?
- (iii) Are the observed numbers of homozygotes and heterozygotes consistent with the assumption that the population from which the reference database was drawn is in Hardy-Weinberg equilibrium?

EXERCISE A.8. Let  $LR = LR_{H_1, H_2}(E)$  be the likelihood ratio in favour of  $H_1$  versus  $H_2$  for evidence  $E$ . You may suppose that the set of obtainable evidence is finite. We assume that  $\mathbb{P}(E | H_1) > 0$  implies that  $\mathbb{P}(E | H_2) > 0$ , for every possible evidence  $E$ . Show that for all  $a, b$  such that  $\mathbb{P}(LR \in [a, b] | H_1) > 0$ ,

$$\mathbb{P}(LR \in [a, b] | H_2) = \mathbb{E}[LR^{-1} | LR \in [a, b], H_1] \mathbb{P}(LR \in [a, b] | H_1).$$

EXERCISE A.9. Suppose that at the scene of a crime, some DNA is found, and that there is a suspect  $S$  who is suspected to be the source of that DNA. Assume that we know that  $S$  has the same DNA profile as the one found at the scene of the crime. Consider the following hypotheses:

- $H_p$ : The suspect  $S$  is the source of the DNA.
- $H_d$ : A person unrelated to the suspect  $S$  is the source of the DNA found.

A forensic expert claims that the likelihood ratio of the DNA match for  $H_p$  versus  $H_d$  is 10 million, and we may assume that the expert made no mistake.

- (i) Can we conclude that  $H_p$  is 10 million times more likely than  $H_d$ ?
- (ii) Can we conclude that  $H_p$  explains the evidence better than  $H_d$ ?
- (iii) Do you agree with the statement that the results provide strong evidence in favour of  $H_p$ ?

SOLUTION. (i) No. That is a statement about the posterior probability, the likelihood ratio is not, and we know nothing about the prior odds.

- (ii) Yes. That is precisely what the likelihood ratio says: the results are 10 million times more likely if  $H_p$  is true than if  $H_d$  is true.
- (iii) That depends on which other hypotheses can also play a role. If the suspect's brother could also be a contributor, the evidence will be weaker when  $H_p$  is compared to that possibility. However,  $H_p$  is the best explanation possible since we have a full match; so (excepting monozygotic twins) the match does provide evidence for  $H_p$  compared to all alternatives.

□

EXERCISE A.10. Consider a situation in which the mother  $M$  of a child  $C$  is known. The father is assumed to be one of  $N + 1$  men who together form the collection of possible fathers  $X$ . We assume that we have typed DNA profiles of both mother and child. Now we consider a particular man  $S \in X$ , type his DNA profile, and ask what the probability is that  $S$  is the father of  $C$ . We write  $F$  for the event that  $S$  is the father.

We consider the situation in which the DNA profiles comprise only one locus, and we assume that allele frequencies are known. We write  $G_M$ ,  $G_C$ , and  $G_S$  for the genotype at this particular locus of  $M$ ,  $C$ , and  $S$ , respectively. Suppose now that  $G_M = (a, b)$ .

- (i) Suppose that  $G_C = (a, b)$ . Give expressions for all possible values of  $LR_{F,C}$  and explain when these values are attained.
- (ii) A critic remarks that this procedure must be fallacious since the probability that  $S$  is the father should also depend on the genotypes of the other persons in  $X$ . What is your reply to this critic?

EXERCISE A.11. A recently published article on the interpretation of DNA mixture profiles states the following.

*Even if [...] a person's alleles are included on every locus of a mixed DNA profile then, while there will be a very high LR, the probability that the person is a contributor (in the absence of any other evidence) is actually very low. This is because, as we have explained above, there are many combinations of profiles that do not include the suspect's. Hence, in the absence of any other evidence, the probability for the defendant's inclusion in a mixed profile will be exceptionally low.*

You may assume that the mixed profiles that are referred to, consist of the set of all alleles of all contributors (no artefacts such as drop-in). Briefly indicate whether and why you agree with each of the following parts:

- (i) *If [...] a person's alleles are included on every locus of a mixed DNA profile then, [...] there will be a very high LR.*
- (ii) *There are many combinations of profiles that do not include the suspect's.*
- (iii) *[...] the probability that the person is a contributor (in the absence of any other evidence) is actually very low.*

EXERCISE A.12. Consider a collection  $X$  of  $N + 1$  individuals who together form the inhabitants of an island. On this island a crime is committed by an inhabitant. The (unknown) criminal is denoted by  $C$ . Furthermore, we have a DNA profile  $\Gamma$  found at the scene of the crime. For  $x \in X$ , we write  $\Gamma_x = 1$  if  $x \in X$  has the characteristic  $\Gamma$  and  $\Gamma_x = 0$  otherwise. The  $\Gamma_x$  are *i.i.d.* Bernoulli distributed with success probability  $p$  and are also independent of  $C$ . We write  $E_C$  for the event that  $\Gamma_C = 1$  and  $U$  for the number of  $\Gamma$ -bearers. Conditional on  $E_C$ ,  $U$  is distributed as 1 plus a binomial distribution with parameters  $N$  and  $p$ . We check randomly chosen inhabitants of the island one by one until we find someone with the characteristic  $\Gamma$ . We call this person  $S$ .

(i) Argue that

$$\mathbb{P}(S = C \mid E_C) = \sum_{k=1}^{N+1} \frac{1}{k} \mathbb{P}(U = k \mid E_C).$$

(ii) Show by explicit computation of the expression in (i) that

$$\mathbb{P}(S = C \mid E_C) = \frac{1 - (1 - p)^{N+1}}{p(N + 1)}.$$

Now suppose that we are uncertain about the value of  $p$ . There is a random variable  $W$ , taking values in  $[0, 1]$  with probability density  $\chi$ , so that conditional on  $W = r$ , the  $\Gamma_x$  are independent Bernoulli random variables with  $\mathbb{P}(\Gamma_x = 1) = r$ . We assume that  $\mathbb{E}(W) = p$ .

(iii) Show that the density  $\chi_C(t)$  of  $W$  conditioned on  $E_C$  is given by  $\chi_C(t) = \frac{t}{p} \chi(t)$ .

(iv) Argue that the expression in (i) is still valid, and compute the expression by conditioning on  $W$ .

EXERCISE A.13. Suppose that a man whom we call  $AF$  (Alleged Father) is considered a possible father of a child  $C$ . It turns out that  $AF$  and  $C$  have at least one allele in common at all considered loci. Thus,  $AF$  cannot be ruled out as a possible father. The result of a paternity test can be reported in two different ways:<sup>1</sup>

- From the child's DNA profile we compute the probability that a random man, unrelated to the child, cannot be excluded as the father of the child since he shares at least one allele with the child at every locus. Call this probability RMNE (Random Man Not Excluded). Note that this probability depends only on the genotype of the child.
- By reporting the likelihood ratio

$$\text{LR}(AF) = \frac{\mathbb{P}(G_{AF}, G_C \mid H_1)}{\mathbb{P}(G_{AF}, G_C \mid H_2)},$$

where  $G_{AF}, G_C$  denote the genotypes of  $AF$  and  $C$  respectively, and where  $H_1$  states that  $AF$  is the biological father of  $C$  and  $H_2$  that  $AF$  is unrelated to  $C$ .

- (i) Suppose a child has genotype  $G_C = (a_1, a_2)$  at one autosomal locus and  $(b_1, b_1)$  at another autosomal locus. Let the allele frequencies be  $p_1, p_2$  for  $a_1, a_2$  and  $q_1$  for  $b_1$ . We consider these to be known without uncertainty. What is the RMNE for each locus separately, and for these loci together?

<sup>1</sup> We ignore the possibility of mutation in both approaches.

- (ii) Give, considering one autosomal locus, an example of genotype  $G_{AF}, G_C$  for  $AF$  and a choice of allele frequencies such that  $LR(G_{AF}, G_C) > \frac{1}{RMNE}$  and also a choice such that  $LR(G_{AF}, G_C) < \frac{1}{RMNE}$ .
- (iii) Show that  $\frac{1}{RMNE}$  can also be interpreted as a likelihood ratio for hypotheses  $H_1$  versus  $H_2$ , by identifying the evidence this likelihood ratio takes into account.
- (iv) Which method do you prefer, and why?

SOLUTION. (i) A random man is excluded as the father at the first locus if and only if neither of his alleles is  $a_1$  or  $a_2$ , which happens with probability  $(1 - (p_1 + p_2))^2$ . Thus, the RMNE probability at the first locus is

$$1 - (1 - (p_1 + p_2))^2 = 2(p_1 + p_2) - (p_1 + p_2)^2.$$

At the second locus, we analogously obtain  $1 - (1 - q_1)^2 = 2q_1 - q_1^2$ . For autosomal loci, we can multiply these probabilities to obtain the RMNE for both loci together.

- (ii) For example, let  $G_{AF} = (a_1, a_3)$ ,  $G_C = (a_1, a_2)$ . Then

$$LR = \frac{\mathbb{P}(G_C | G_{AF}, H_1)}{\mathbb{P}(G_C | H_2)} = \frac{\frac{1}{2}p_2}{2p_1p_2} = \frac{1}{4p_1},$$

$$\frac{1}{RMNE} = \frac{1}{2(p_1 + p_2) - (p_1 + p_2)^2}.$$

Choose a small value of  $p_1$ , for example  $p_1 = \frac{1}{400}$ ; then  $LR = 100$ . With a large value of  $p_2$ , RMNE is large. For example, if  $p_2 = \frac{199}{400}$ , then  $p_1 + p_2 = \frac{1}{2}$  and  $\frac{1}{RMNE} = \frac{4}{3} < LR$ . With the same frequencies, now consider  $G_{AF} = (a_2, a_3)$ ; then

$$LR = \frac{1}{4p_2} = \frac{100}{199} < \frac{1}{RMNE}.$$

- (iii) If we take as evidence  $E$  that  $AF$  has an allele in common with  $C$  on all loci, then  $\mathbb{P}(E | H_1, G_C) = 1$ ,  $\mathbb{P}(E | H_2, G_C) = RMNE$ .
- (iv) The likelihood ratio, since it takes more information into account: it also takes the genotype  $G_{AF}$  into account and RMNE does not.

□

EXERCISE A.14. The 2016 *Report by the US President's Council of Advisors on Science and Technology* (PCAST) made a number of recommendations for the future development of forensic science. On page 114 of that report we read: “Footwear analysis is a process that typically involves comparing a known object, such as a shoe, to a complete or partial impression found at a crime scene, to assess whether the object is likely to be the source of the impression.” Comment on this statement.

SOLUTION. One cannot assess how likely it is that the object is the source of the impression: that is given by the posterior odds/probabilities. These depend on the prior odds which are obtained from the context that is not always fully known by the forensic lab. The forensic analysis consists of assigning a likelihood ratio. □

EXERCISE A.15. Consider a situation in which the mother  $M$  of a child  $C$  is known. The father is assumed to be one of  $N + 1$  men who together form the collection of possible fathers  $X$ . We assume that we have typed DNA profiles of both mother and child at one locus. Suppose that  $M$  has genotype  $(a, b)$  and  $C$  has genotype  $(b, c)$ , where  $a, b$ , and  $c$  are all different and have population frequencies  $p_a, p_b$ , and  $p_c$ , respectively. For every  $x \in X$ , let  $\Gamma_x = 1$  if  $x$  has at least one allele  $c$  at the locus under consideration.

- (i) What is the probability that a person  $y$ , unrelated to  $C$ , has  $\Gamma_y = 1$ ?

- (ii) An islander  $S$  is selected and turns out to have  $\Gamma_S = 1$ . What is the posterior probability that  $S$  is the true father, assuming uniform priors?
- (iii) Now we learn the actual number of alleles  $c$  of  $S$ . What is the posterior probability that  $S$  is the father if  $S$  has only one allele  $c$ ? And when he has two alleles  $c$ ?
- (iv) Elaborate on the interpretation of probability in forensic science given the answers that you gave in (ii) and (iii).

SOLUTION. (i) Such a person must have at least one allele  $c$ , and this is so with probability  $2p_c(1 - p_c) + p_c^2 = 2p_c - p_c^2 = p_c(2 - p_c)$ .

- (ii) Let  $H_1$  denote the hypothesis that  $S$  is the father of  $C$ , and  $H_2$  the hypothesis that  $S$  is unrelated to  $C$ . Uniform priors imply

$$\mathbb{P}(H_1) = \frac{1}{N+1}, \quad \mathbb{P}(H_2) = \frac{N}{N+1}.$$

Then

$$\frac{\mathbb{P}(H_1 | E)}{\mathbb{P}(H_2 | E)} = \frac{1}{p_c(2 - p_c)} \cdot \frac{\frac{1}{N+1}}{\frac{N}{N+1}} = \frac{1}{Np_c(2 - p_c)}.$$

The posterior probability is

$$\mathbb{P}(H_1 | E) = \frac{1}{1 + Np_c(2 - p_c)}.$$

- (iii) If  $S$  has one allele  $c$ , suppose his genotype is  $(c, x)$ . The likelihood ratio in favour of paternity is then

$$\frac{\mathbb{P}(G_C = (b, c) | G_M = (a, b), G_S = (c, x), H_1)}{\mathbb{P}(G_C = (b, c) | G_M = (a, b), G_S = (c, x), H_2)} = \frac{\frac{1}{4}}{\frac{1}{2}p_c} = \frac{1}{2p_c}.$$

Therefore the posterior odds are  $\frac{1}{2Np_c}$  and the posterior probability of paternity is  $\frac{1}{1+2Np_c}$ .

If  $S$  has two alleles  $c$ , then his genotype is  $(c, c)$ . The likelihood ratio in favour of paternity is then

$$\frac{\mathbb{P}(G_C = (b, c) | G_M = (a, b), G_S = (c, c), H_1)}{\mathbb{P}(G_C = (b, c) | G_M = (a, b), G_S = (c, c), H_2)} = \frac{\frac{1}{2}}{\frac{1}{2}p_c} = \frac{1}{p_c}.$$

Therefore the posterior odds are  $\frac{1}{Np_c}$  and the posterior probability of paternity is  $\frac{1}{1+Np_c}$ .

- (iv) We see that, if  $S$  has one allele  $c$ , the posterior probability of paternity is

$$\frac{1}{1 + 2Np_c} < \frac{1}{1 + Np_c(2 - p_c)}.$$

If  $S$  has two alleles  $c$ , the posterior probability of paternity is

$$\frac{1}{1 + Np_c} > \frac{1}{1 + Np_c(2 - p_c)}.$$

This is in line with the epistemic interpretation of probability: if we have more knowledge about the genotype of  $S$  (i.e., knowing the number of  $c$ -alleles instead of only knowing that  $S$  has at least one) we are able to better discriminate between paternity and unrelatedness.

□

EXERCISE A.16. Let  $\text{Bel}$  be a belief function and let  $H$  be a hypothesis of interest, say the hypothesis that the suspect is guilty of a certain crime. Define

$$BR_1 := \frac{\text{Bel}(H)}{\text{Bel}(H^c)}.$$

- (i) Suppose that  $BR_1$  is very high. Does this imply that the belief in  $H$  is also high?

- (ii) Do you think that a verdict should be based on the value of  $BR_1$ ?

EXERCISE A.17 (Bayesian Networks). John is found dead in his home, lying on the kitchen floor in a pool of blood, with a severe stab wound. In a garbage container down the road a bloody knife is found. The knife blade is sampled and tests positive for human blood. The DNA profile of this sample matches John's DNA profile and shows no additional alleles. The random match probability is 1 in a billion ( $10^{-9}$ ). The handle of the knife is also sampled. This sample yields a DNA mixture profile, assumed to originate from three donors, matching John and two unknown men. This mixture profile is used to search the DNA database. A match is found with Pete, who then becomes the suspect in this case. The source-level LR of the DNA evidence against Pete is 1,000,000. When his home is searched, a pair of shiny new trainers is found in Pete's indoor garbage bin. The soles of the trainers are compared with a set of bloody shoe marks found next to John's body, and the comparison shows matching patterns. Based on a police database of suspects' footwear, it is assumed that 1 in 800 criminals have shoes with matching patterns. The shoes test negative for blood. However, the test is not perfect: if blood is present, there is a probability of 95% that the test is positive.

It turns out that Pete and John are members of the same running club. The other five members testify that Pete and John had a fight when they were doing the dishes after a dinner party at John's house three days ago. The fight was over the new trainers that John had purchased for the whole club and that were handed out to them that night. Pete disliked the new trainers and refused to pay for them.

- (i) Define the pieces of forensic evidence.
- (ii) Define the relevant source-level hypotheses that are addressed by the forensic evidence.
- (iii) Define the relevant activity-level hypotheses that are addressed by the forensic evidence.
- (iv) Construct a Bayesian network suitable for deriving the combined evidential strength of the forensic findings concerning the knife.
- (v) Construct a Bayesian network suitable for deriving the combined evidential strength of the forensic findings concerning the trainers.
- (vi) Construct a Bayesian network suitable for deriving the combined evidential strength of all forensic findings.
- (vii) Formulate a conclusion about the combined evidential strength of all forensic findings. Explicitly state the hypotheses that are considered.

SOLUTION. (i) We define the pieces of forensic evidence as:

- $E_1$ : The DNA profile of the blood on the knife matches John.
- $E_2$ : The DNA mixture profile of the knife handle matches John, Pete, and one other person.
- $E_3$ : Bloody shoe marks match the soles of the trainers found in Pete's garbage bin.
- $E_4$ : The blood test on the shoes is negative.

- (ii) The source-level hypotheses are:

- $H_{1,p}$ : Part of the DNA profile from the knife handle is from Pete.
- $H_{1,d}$ : The DNA profile from the knife handle is from John and two unknown men.
- $H_{2,p}$ : The bloody shoe marks are from Pete's trainers.
- $H_{2,d}$ : The bloody shoe marks are from an unknown person's shoes.

- (iii) The activity-level hypotheses are:

- $H_{3,p}$ : Pete stabbed John during a fight at John's house.
- $H_{3,d}$ : An unknown man stabbed John during a fight at John's house.
- $H_{4,p}$ : Pete walked through the pool of blood.
- $H_{4,d}$ : An unknown man walked through the pool of blood.

The Bayesian networks for parts (iv)–(vi) are presented in [Figures A.1 to A.3](#), respectively.

- (iv) We first define the hypothesis node, which we call 'Offender?', to represent whether Pete is the offender ( $H_p$ ) or not ( $H_d$ ). This node thus has the two states 'yes' and 'no', corresponding to whether Pete is or is not the offender, respectively. Then we have two situation nodes: 'Pete killed John' and 'An unknown person killed John'. Only one of these situations can occur, and they both have the two states 'yes' and 'no'.

Unless explicitly stated otherwise, all remaining nodes also have the two states 'yes' and 'no'. We have the two transfer nodes: 'Transfer of DNA from Pete to the knife' and 'Transfer of DNA from the unknown person to the knife'. These two nodes are in the state 'yes' if there was such a transfer, and otherwise are in the state 'no'. We include persistence and recovery in these nodes, instead of having separate transfer nodes.

We have the node 'The knife is used by the offender when killing John'. This is a node for whether the offender used the bloody knife that was found in a garbage container down the road, or used something else when killing John. We assume that this node is independent of both situation nodes, since we do not know which states the transfer nodes are in.

Next, we have three nodes for whether or not there is DNA from a specific person on the knife: 'DNA from John on the knife', 'DNA from Pete on the knife', and 'DNA from the unknown person on the knife'. The node 'DNA from John on the knife' can occur at the same time as either of the other nodes, but the other two nodes cannot occur at the same time, since we assumed only one of the two situation nodes can occur at the same time.

The node 'DNA from Pete on the knife' is also influenced by the background node 'Pete touched the knife at the dinner party'. This background node represents whether Pete touched the knife while doing the dishes after the dinner party at John's house three days earlier, when he and John had a fight.

Finally, we have the two evidence nodes: 'The DNA profile of the blood on the knife matches John' and 'DNA profile of the knife handle matches Pete', corresponding to the pieces of evidence  $E_1$  and  $E_2$ , respectively.

We use the random match probability of 1 in a billion in the conditional probability for the evidence node for  $E_1$ . We use the likelihood ratio of  $10^6$  for the DNA evidence  $E_2$  against Pete. Evidence  $E_2$  is a million times less likely when Pete's DNA is absent from the knife than when it is present, and this ratio is incorporated into the conditional probabilities for the evidence node corresponding to  $E_2$ .

All these nodes and the arrows between them are shown in [Figure A.1](#).

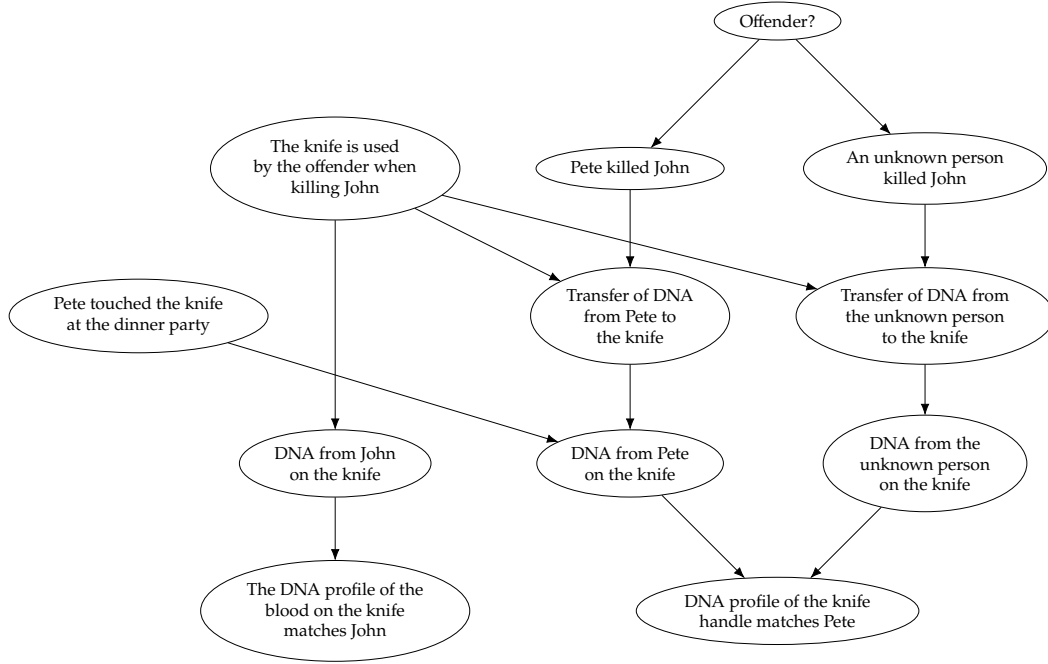


Figure A.1: Bayesian network for the knife.

- (v) As in part (iv), we define the hypothesis node, which we call ‘Offender?’, to represent whether Pete is the offender ( $H_p$ ) or not ( $H_d$ ).

We have the following three situation nodes: ‘Pete killed John wearing his trainers’, ‘Pete killed John wearing his other shoes’, and ‘An unknown person killed John’. Only one of these three situations can occur at the same time, and they all have the two states ‘yes’ and ‘no’. We will also have the two states ‘yes’ and ‘no’ for all the other nodes, except when explicitly stated otherwise. We have three transfer nodes: ‘Transfer of blood from John to Pete’s trainers’, ‘Transfer of blood from John to Pete’s other shoes’, and ‘Transfer of blood from John to an unknown man’s shoes’. These three transfer nodes are in the state ‘yes’ if there was such a transfer, and otherwise are in the state ‘no’. As in part (iv), we also include persistence and recovery in these nodes, instead of having separate nodes.

Then we have two nodes: ‘Blood on Pete’s trainers’ and ‘Bloody shoe marks left at John’s house’, for whether or not there is blood on Pete’s trainers and whether or not the bloody shoe marks left under these three situations match the soles of Pete’s trainers. The node ‘Blood on Pete’s trainers’ can occur at the same time as the node ‘Bloody shoe marks left at John’s house’.

Finally, we have the two evidence nodes: ‘Bloody shoe marks match the soles of Pete’s trainers’ and ‘Blood test on the shoes’, for the pieces of evidence  $E_3$  and  $E_4$ , respectively. The evidence node for  $E_4$  is in the state ‘positive’ if there is blood on the shoes and the test detects it (probability 95%), and in the state ‘negative’ if there is no blood on the shoes or the test gives a false negative (probability 5% when blood is present).

We use the 1-in-800 probability of coincidentally matching shoe patterns in the conditional probabilities for the evidence node corresponding to  $E_3$ . For the evidence node corresponding to  $E_4$ , we use the 95% probability that the test is positive when blood is present.

For the conditional probabilities, we treat the node ‘Bloody shoe marks left at John’s house’ as the latent source of the evidence  $E_3$ . This is because it would not make much sense to condition on the three transfer nodes otherwise. Hence, in Hugin, we used the 1-in-800 probability for the node ‘Bloody shoe marks left at John’s house’, while the evidence node for  $E_3$  has deterministic conditional probabilities (0 or 1).

All these nodes and the arrows between them are shown in [Figure A.2](#).

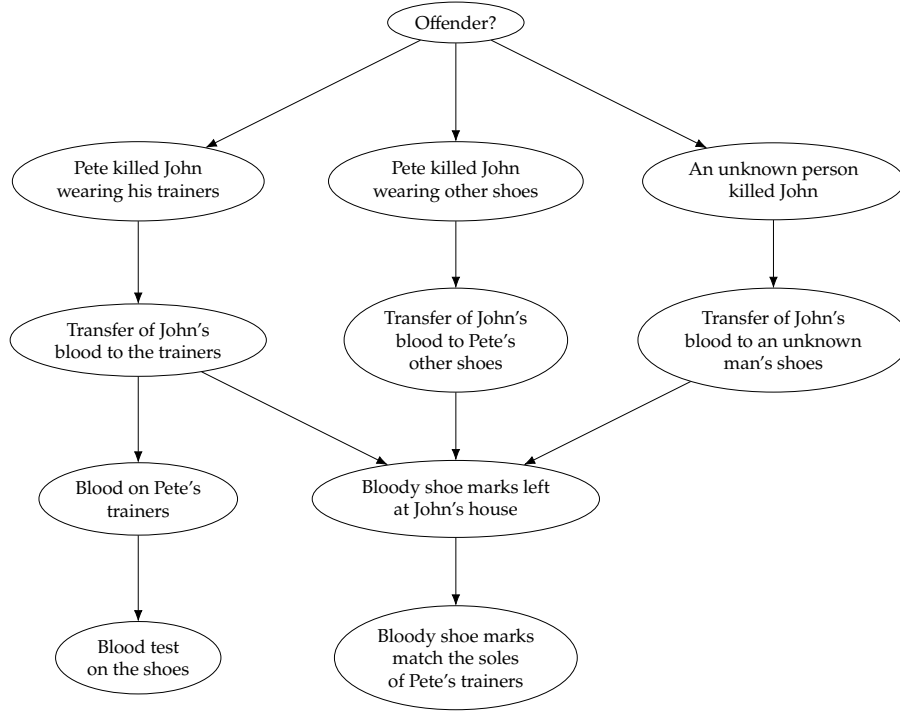


Figure A.2: Bayesian network for the trainers.

- (vi) We combine the two Bayesian networks from parts (iv) and (v) by splitting the situation node 'Pete killed John' from part (iv) into the two situation nodes 'Pete killed John wearing his trainers' and 'Pete killed John wearing other shoes' from part (v). In this way, the networks share the same hypothesis and situation nodes. The situation nodes now have arrows pointing towards nodes from both Bayesian networks, but the remaining nodes from the network in part (iv) do not influence those from the network in part (v), and vice versa.

The resulting Bayesian network is shown in [Figure A.3](#).

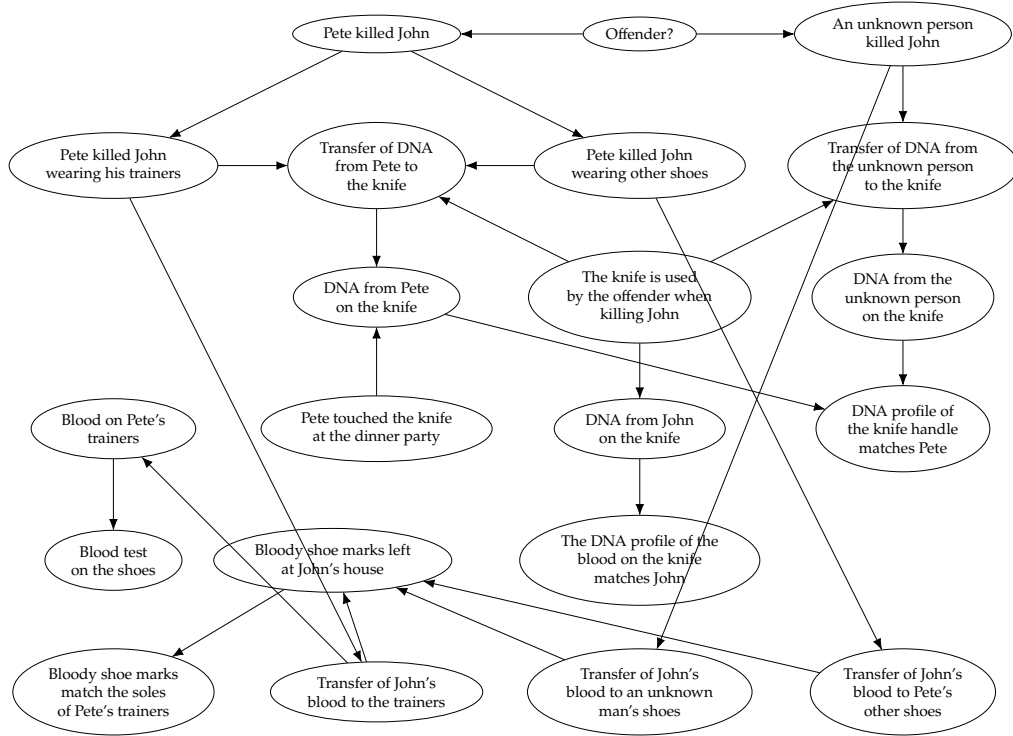


Figure A.3: Bayesian network for all forensic findings.

(vii) We consider the hypotheses:

- $H_p$ : Pete is the offender.
- $H_d$ : An unknown man is the offender.

In the Bayesian network, we considered all four pieces of evidence. The first piece of evidence,  $E_1$ , is neutral with respect to these hypotheses. The evidence  $E_2$  strongly supports  $H_p$  with a likelihood ratio of  $10^6$ . The third piece of evidence,  $E_3$ , also supports  $H_p$  more than  $H_d$ . Finally, the last piece of evidence,  $E_4$ , supports  $H_d$  more than  $H_p$ .

In Hugin, we found that, given all the pieces of evidence, the probability that Pete is the offender is 90.7%.

□

# References

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